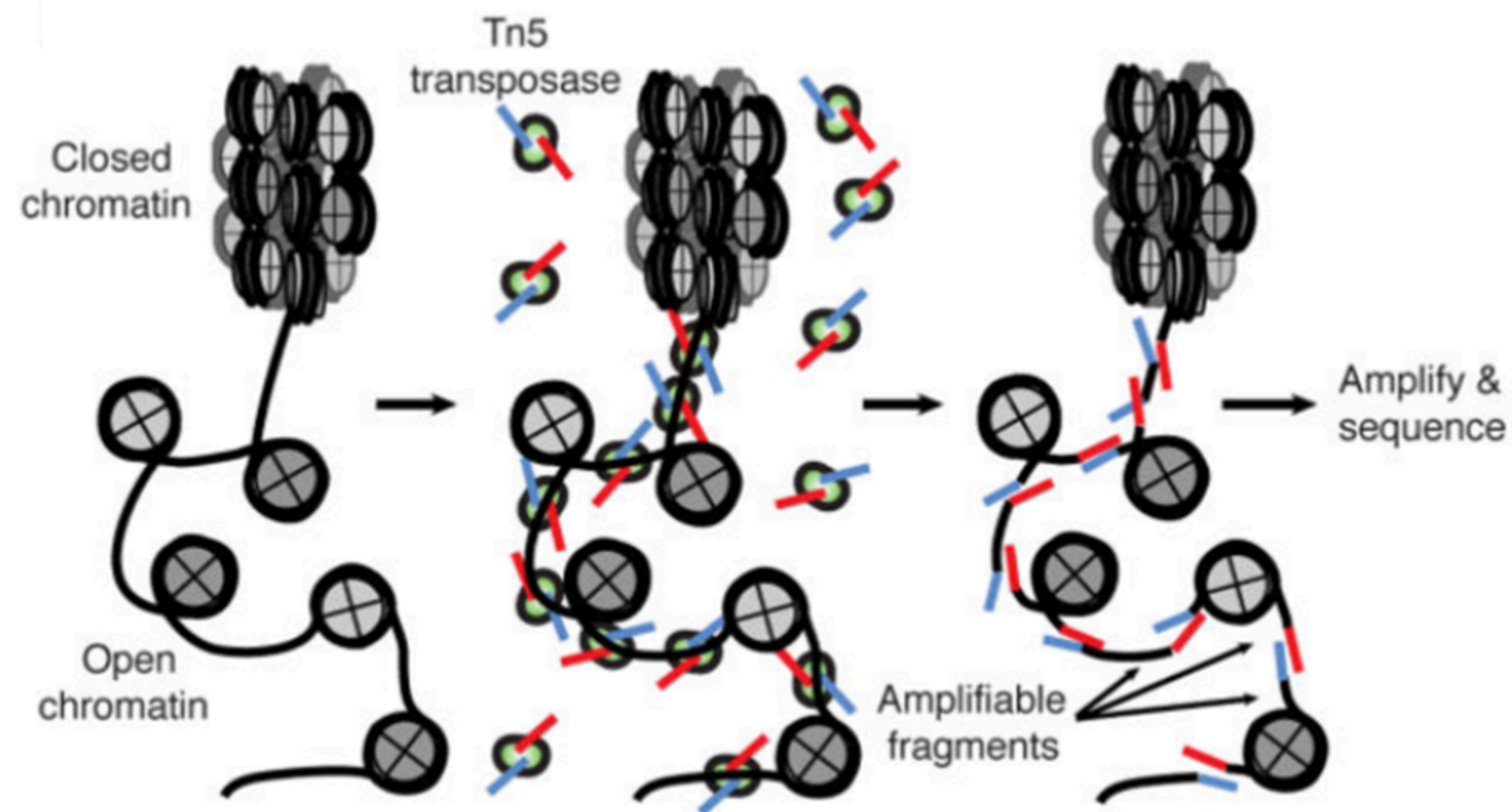


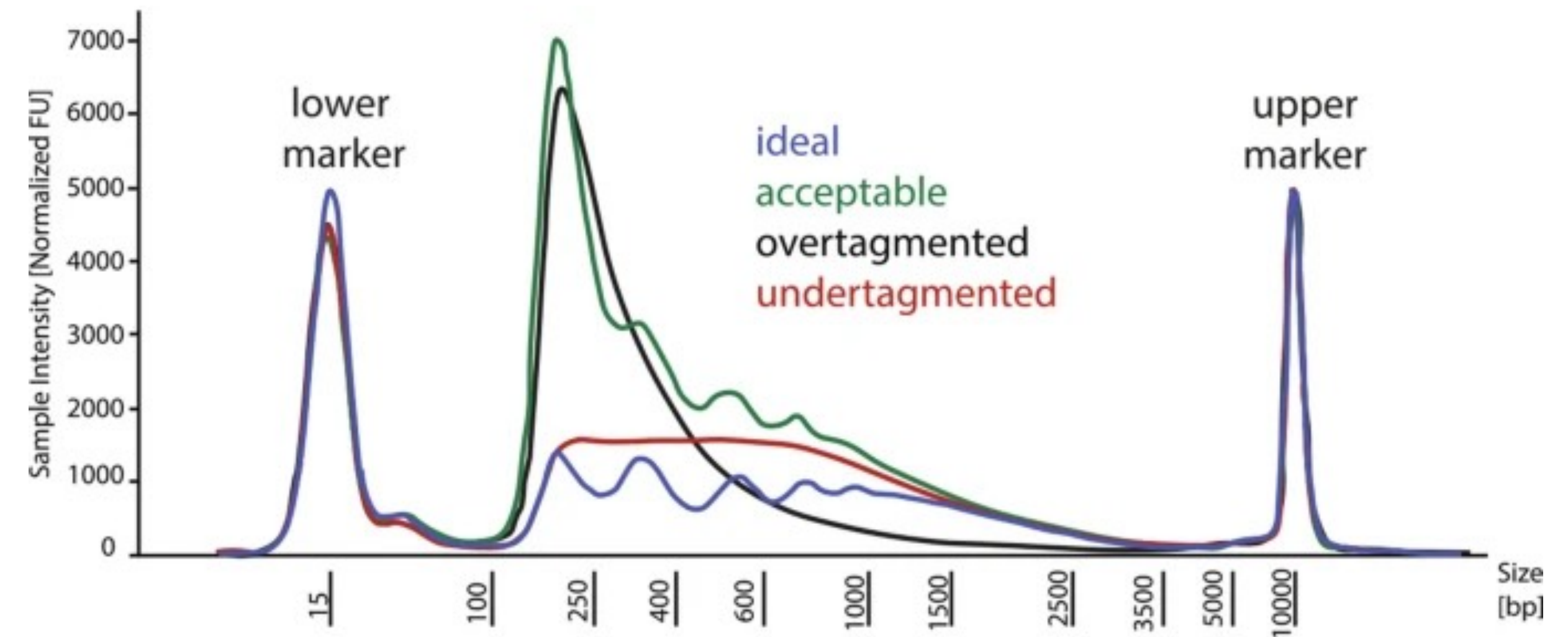
Assay for **T**ransposase **A**ccessible **C**hromatin (ATAC) sequencing

Tagmentation of chromatin by Tn5 transposase



Buenrostro et al., 2015 (Nature Methods)

Fragment size distribution of amplified ATAC-seq library



Gehrke and Srivastava, 2022

FastQC is used to asses the quality of the sequencing reads associated with the samples

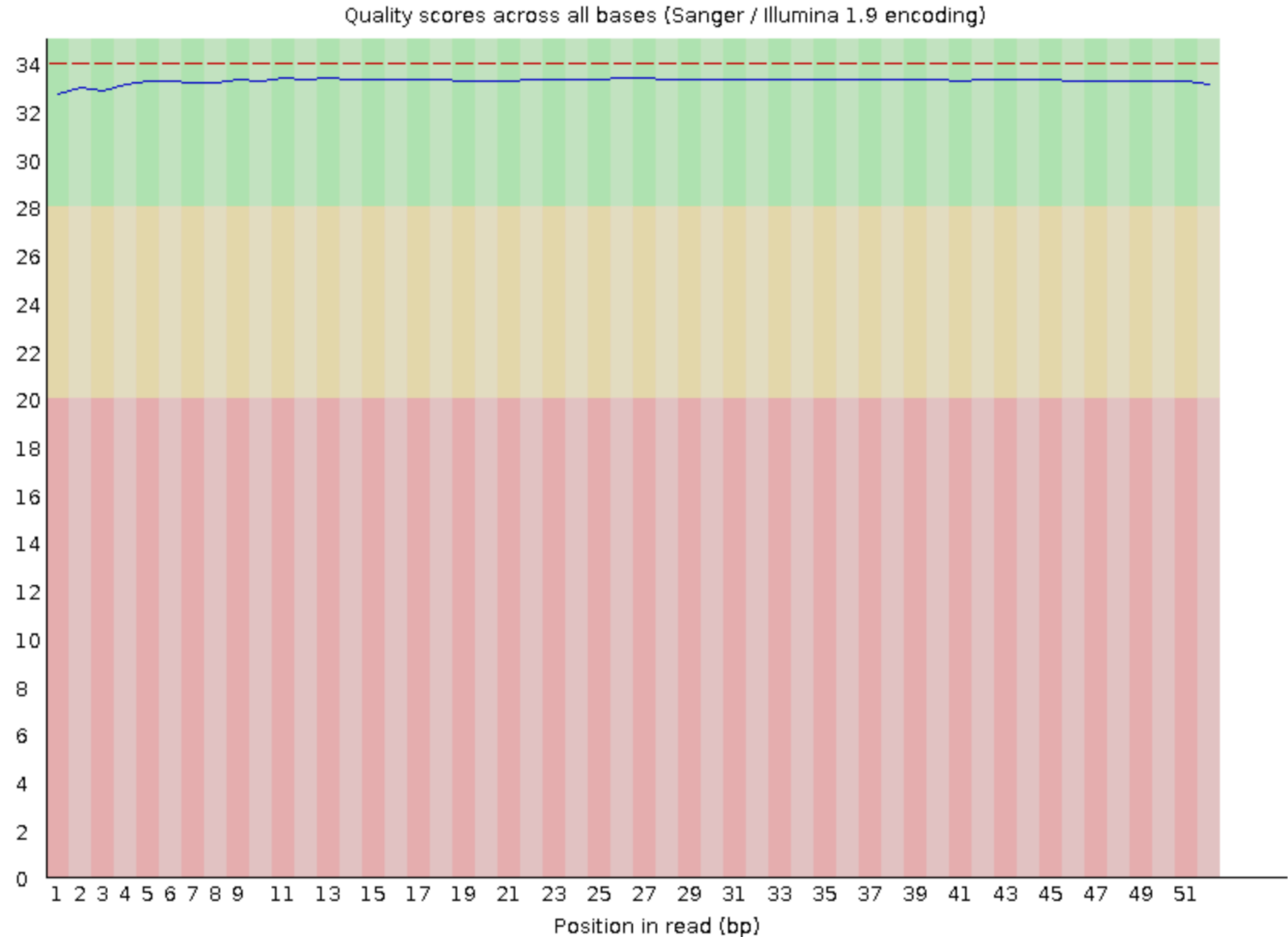
Summary

- ✔ [Basic Statistics](#)
- ✔ [Per base sequence quality](#)
- ✘ [Per tile sequence quality](#)
- ✔ [Per sequence quality scores](#)
- ✘ [Per base sequence content](#)
- ✔ [Per sequence GC content](#)
- ✔ [Per base N content](#)
- ✔ [Sequence Length Distribution](#)
- ✔ [Sequence Duplication Levels](#)
- ✔ [Overrepresented sequences](#)
- ✔ [Adapter Content](#)

✔ Basic Statistics

Measure	Value
Filename	pg_o91_control_1_S7_R1_001.fastq.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	20059216
Sequences flagged as poor quality	0
Sequence length	52
%GC	48

✔ Per base sequence quality



FastQC is used to asses the quality of the sequencing reads associated with the samples

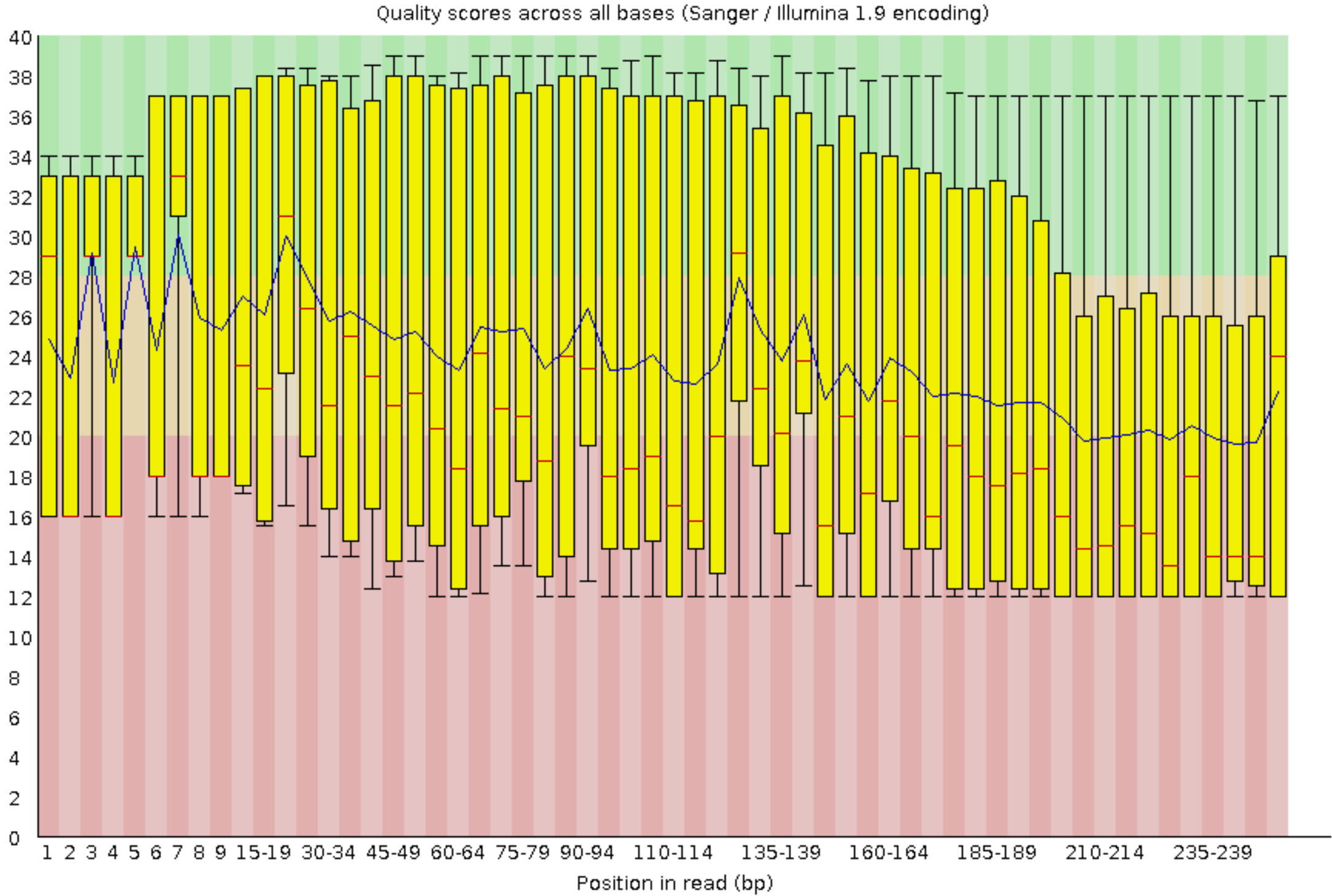
Summary

- ✓ Basic Statistics
- ✓ Per base sequence quality
- ✗ Per tile sequence quality
- ✓ Per sequence quality scores
- ✗ Per base sequence content
- ✓ Per sequence GC content
- ✓ Per base N content
- ✓ Sequence Length Distribution
- ✓ Sequence Duplication Levels
- ✓ Overrepresented sequences
- ✓ Adapter Content

Basic Statistics

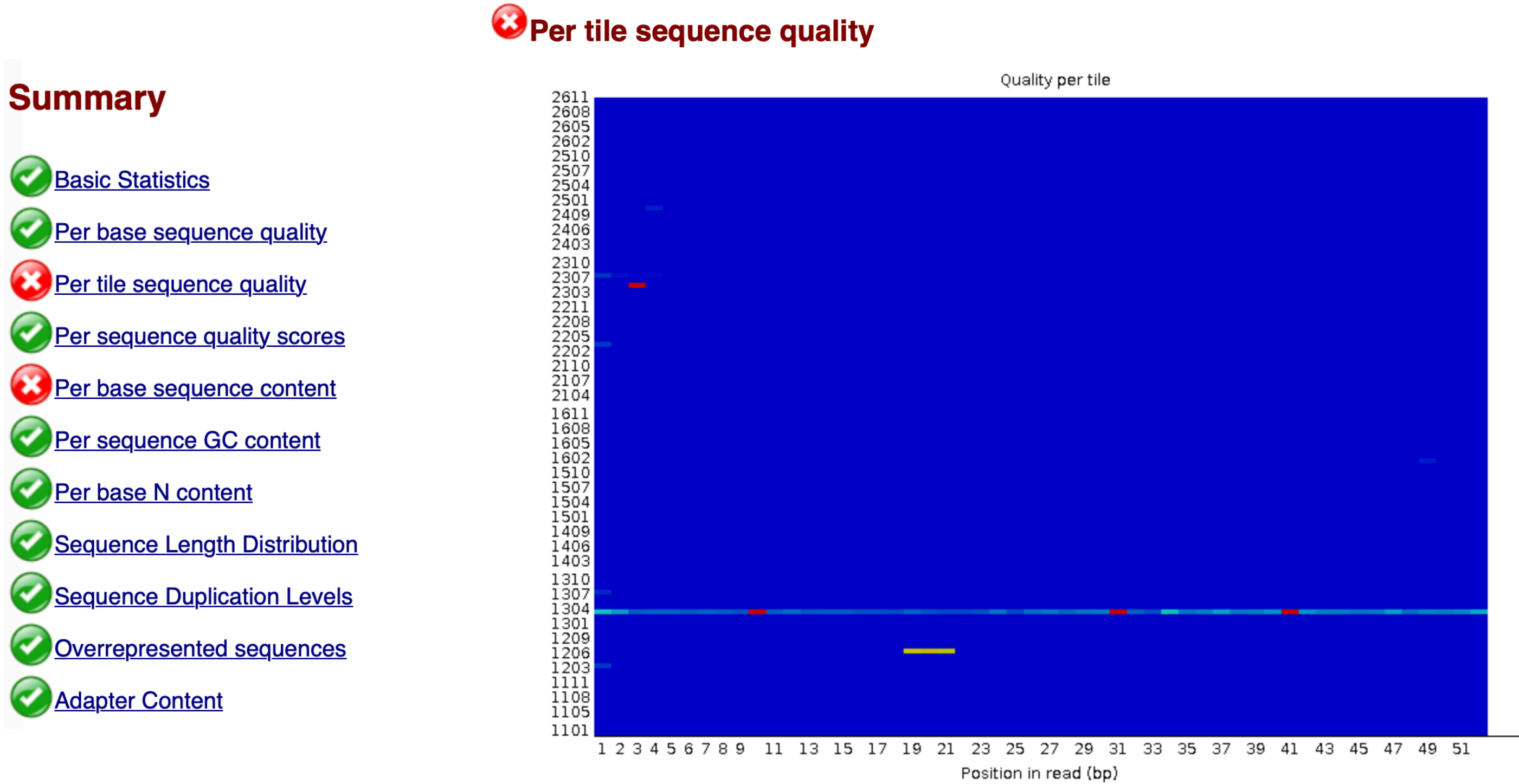
Measure	Value
Filename	pg_o91_control_1_S7_R1_001.fastq.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
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Sequences flagged as poor quality	0
Sequence length	52
%GC	48

✗ Per base sequence quality



<https://www.biostars.org/p/340398/>

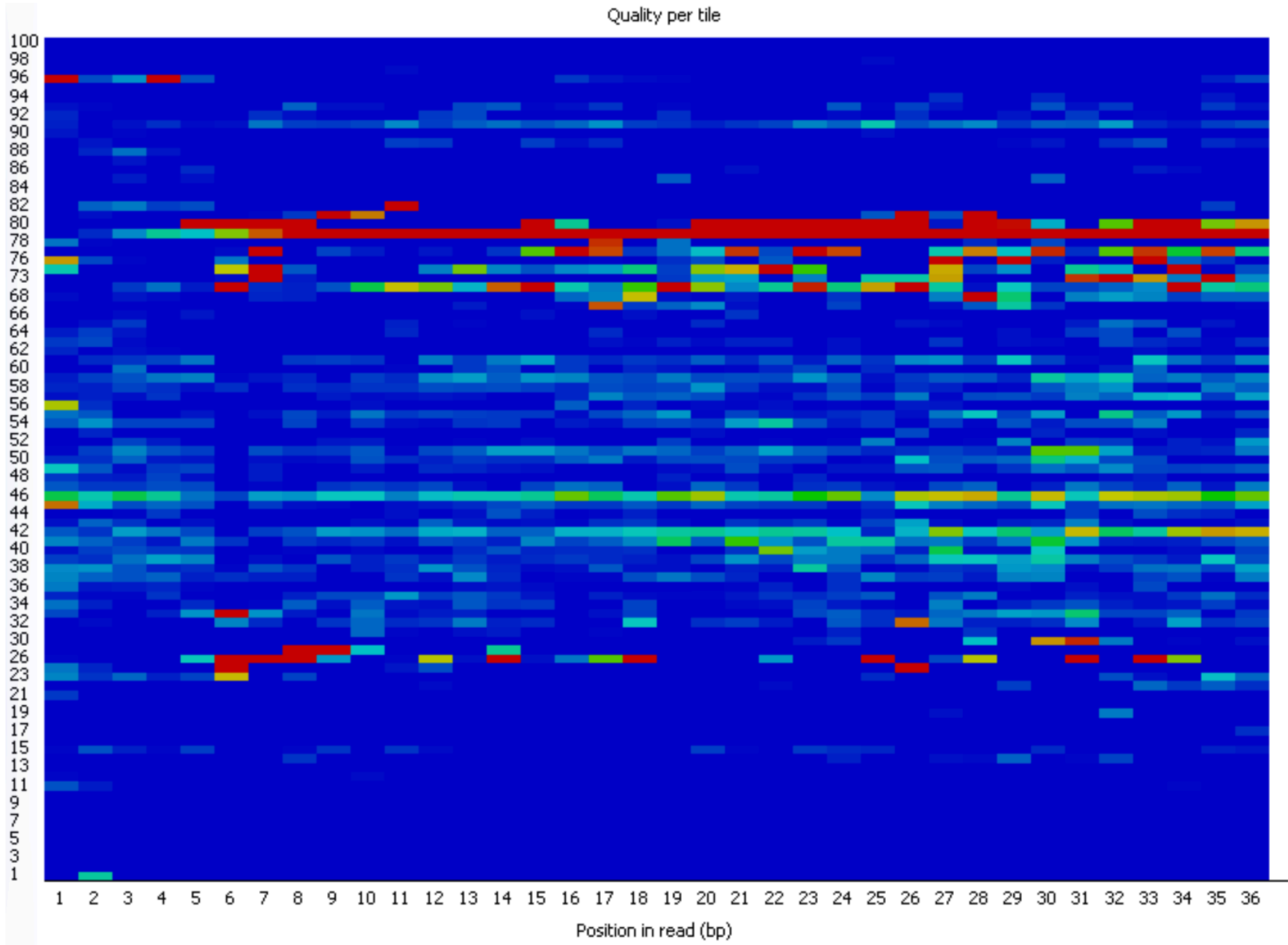
FastQC is used to asses the quality of the sequencing reads associated with the samples



FastQC is used to asses the quality of the sequencing reads associated with the samples

Summary

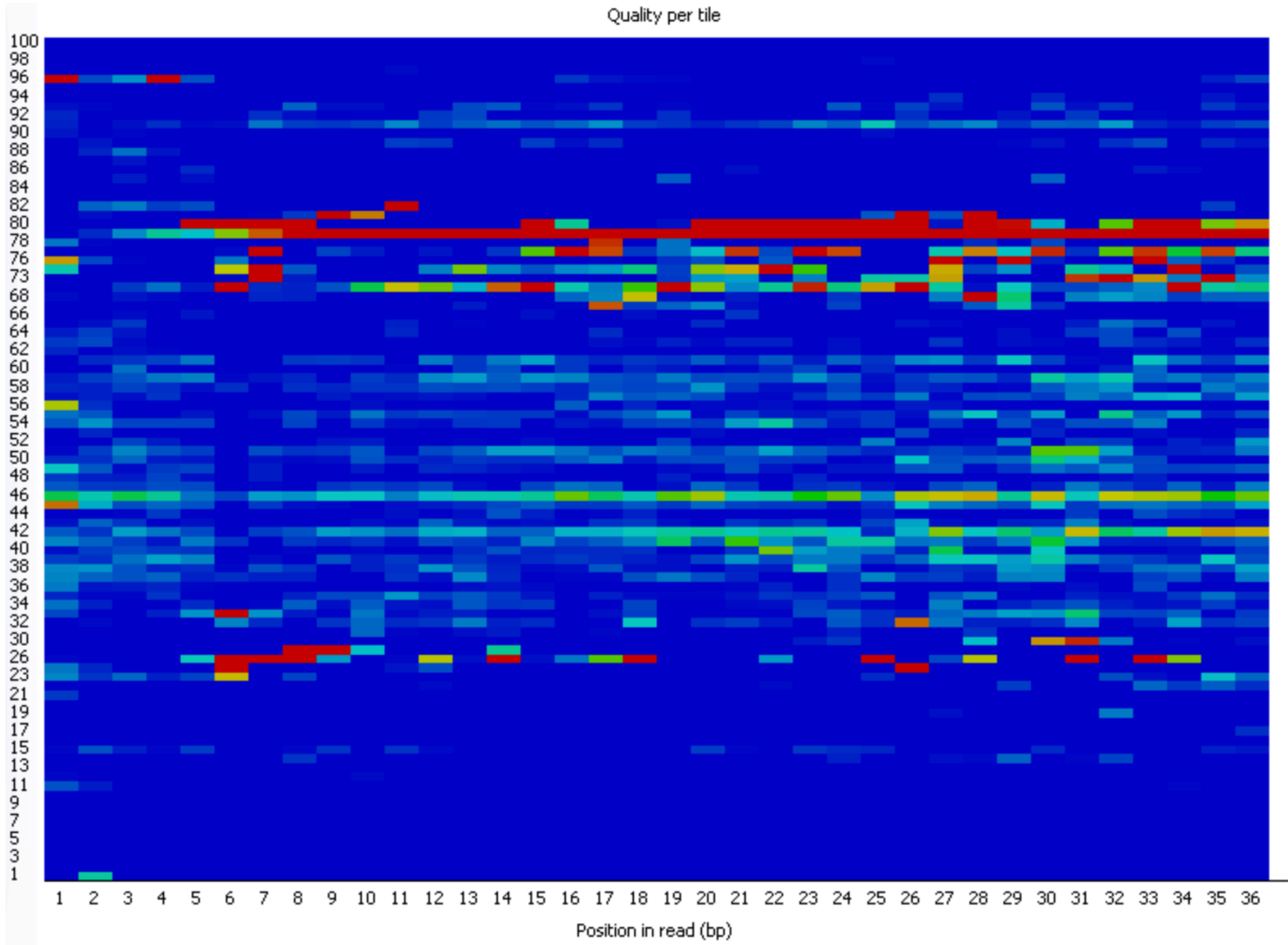
- ✓ [Basic Statistics](#)
- ✓ [Per base sequence quality](#)
- ✗ [Per tile sequence quality](#)
- ✓ [Per sequence quality scores](#)
- ✗ [Per base sequence content](#)
- ✓ [Per sequence GC content](#)
- ✓ [Per base N content](#)
- ✓ [Sequence Length Distribution](#)
- ✓ [Sequence Duplication Levels](#)
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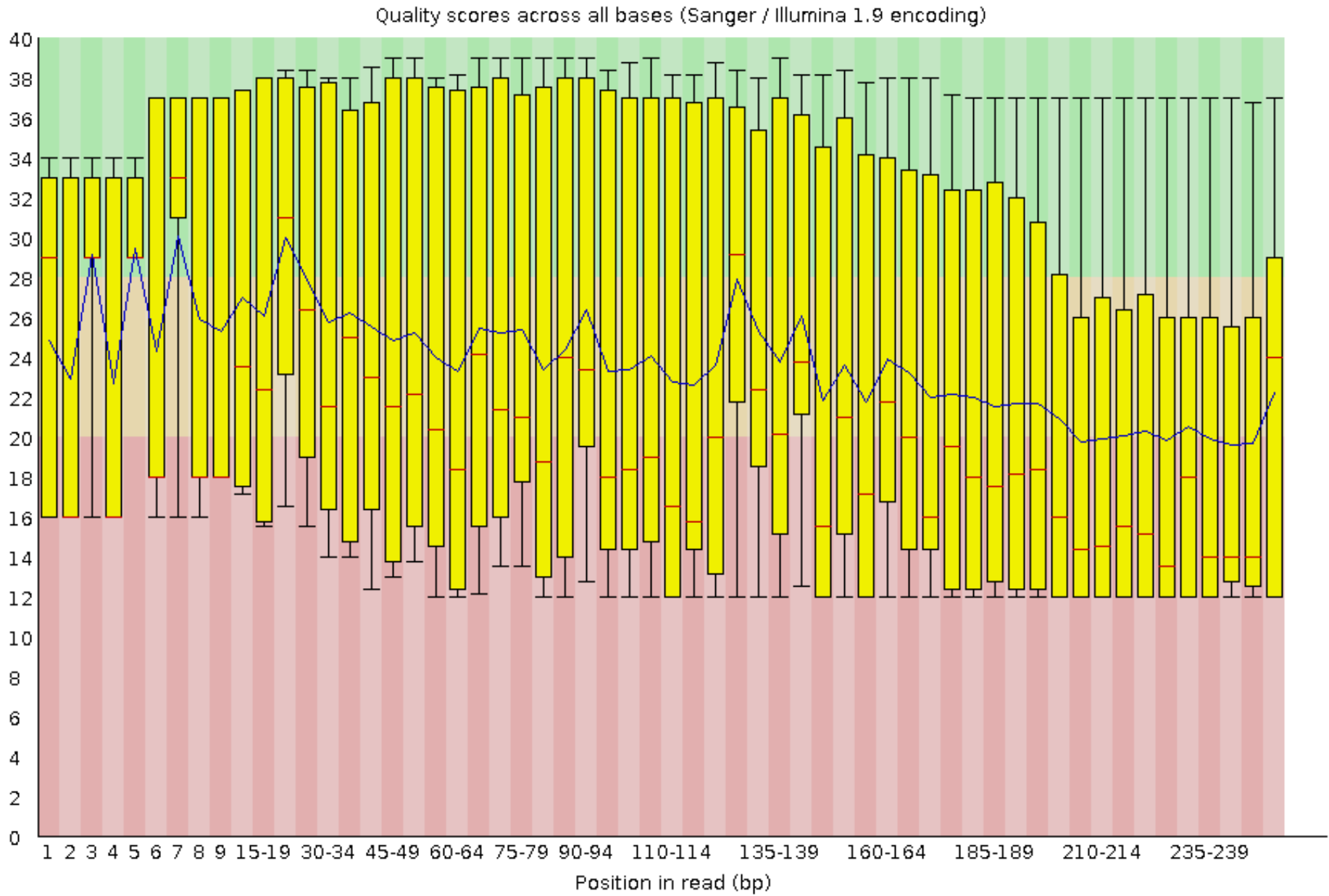
FastQC is used to asses the quality of the sequencing reads associated with the samples

Summary

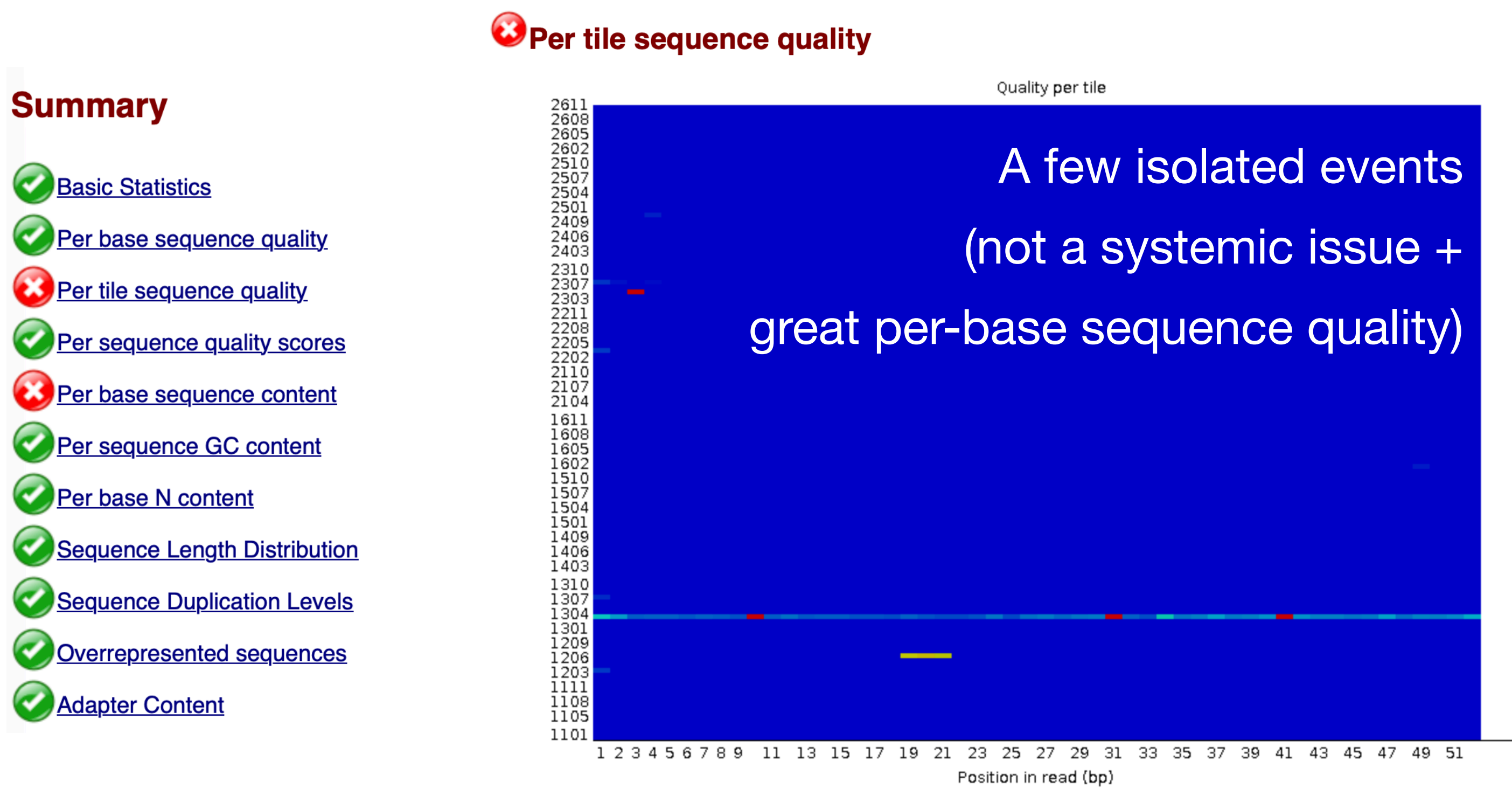
- ✓ [Basic Statistics](#)
- ✓ [Per base sequence quality](#)
- ✗ [Per tile sequence quality](#)
- ✓ [Per sequence quality scores](#)
- ✗ [Per base sequence content](#)
- ✓ [Per sequence GC content](#)
- ✓ [Per base N content](#)
- ✓ [Sequence Length Distribution](#)
- ✓ [Sequence Duplication Levels](#)
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- ✓ [Adapter Content](#)



✗ **Per base sequence quality**



FastQC is used to asses the quality of the sequencing reads associated with the samples

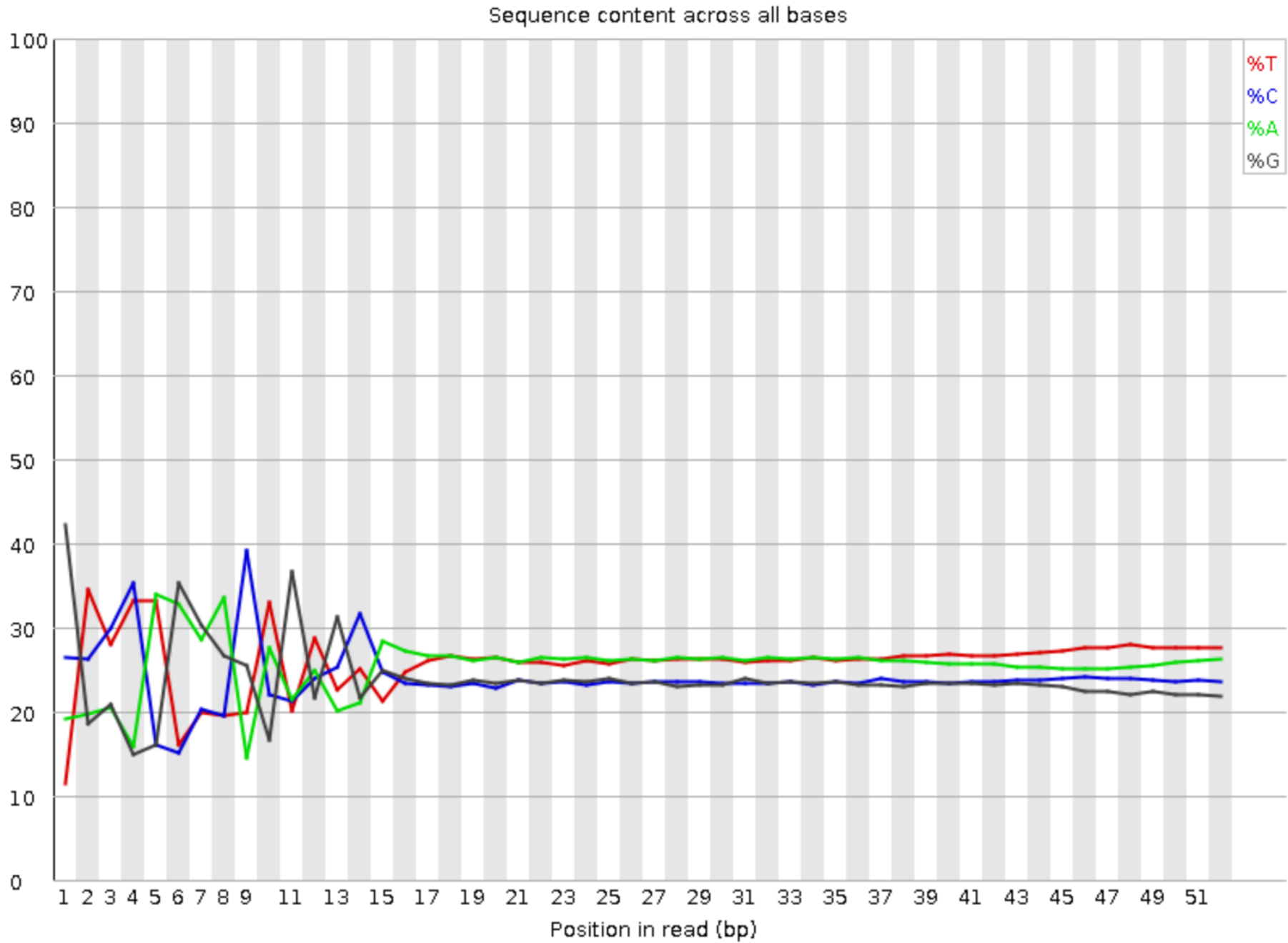


FastQC is used to asses the quality of the sequencing reads associated with the samples

✖ Per base sequence content

Summary

- ✔ [Basic Statistics](#)
- ✔ [Per base sequence quality](#)
- ✖ [Per tile sequence quality](#)
- ✔ [Per sequence quality scores](#)
- ✖ [Per base sequence content](#)
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- ✔ [Adapter Content](#)



Green et al. *Mobile DNA* 2012, **3**:3
<http://www.mobilednajournal.com/content/3/1/3>



SHORT REPORT

Open Access

Insertion site preference of Mu, Tn5, and Tn7 transposons

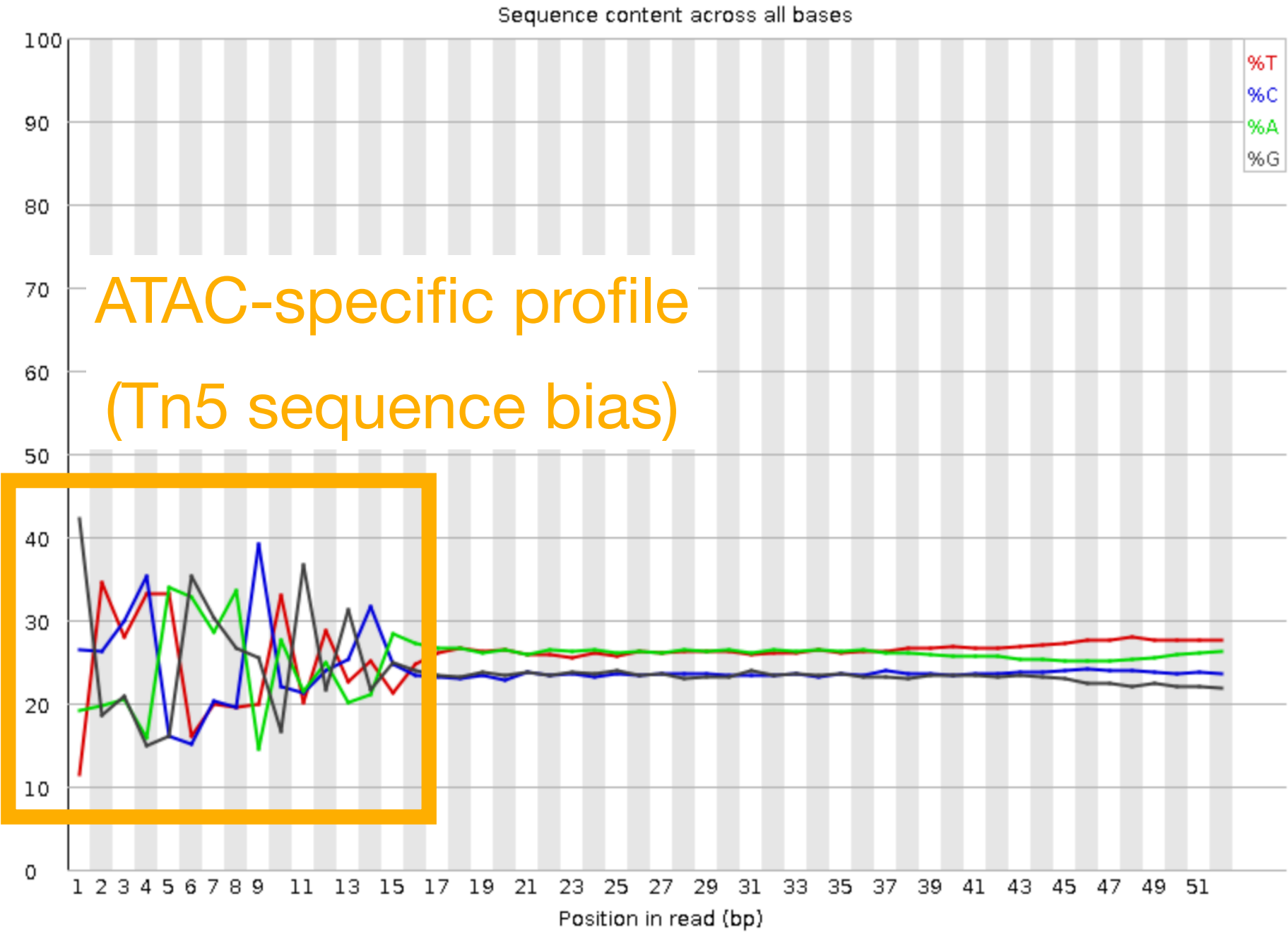
Brian Green¹, Christiane Bouchier², Cécile Fairhead³, Nancy L Craig⁴ and Brendan P Cormack^{1*}

FastQC is used to asses the quality of the sequencing reads associated with the samples

✖ Per base sequence content

Summary

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Green et al. *Mobile DNA* 2012, **3**:3
<http://www.mobilednajournal.com/content/3/1/3>



SHORT REPORT

Open Access

Insertion site preference of Mu, Tn5, and Tn7 transposons

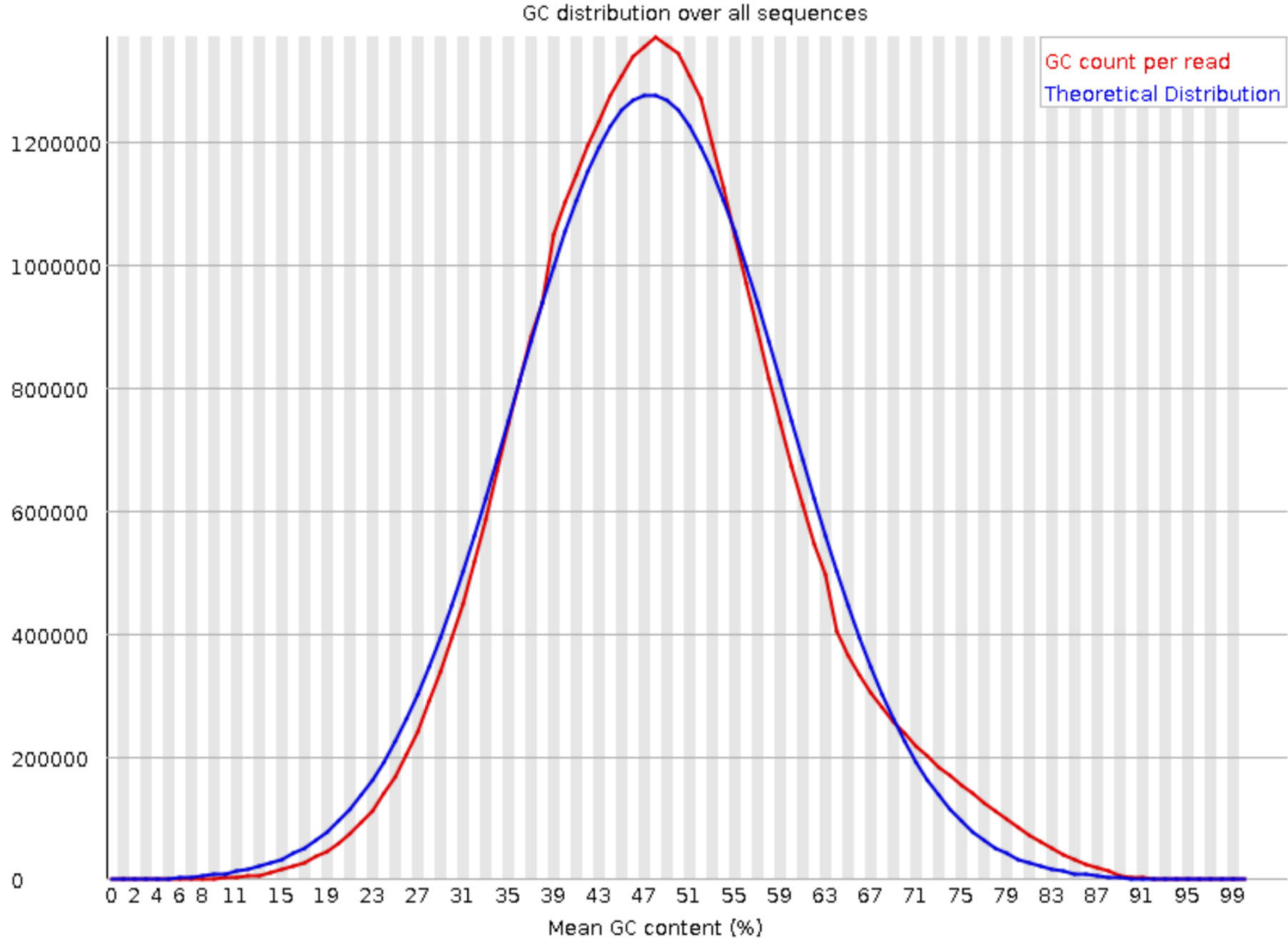
Brian Green¹, Christiane Bouchier², Cécile Fairhead³, Nancy L Craig⁴ and Brendan P Cormack^{1*}

FastQC is used to asses the quality of the sequencing reads associated with the samples

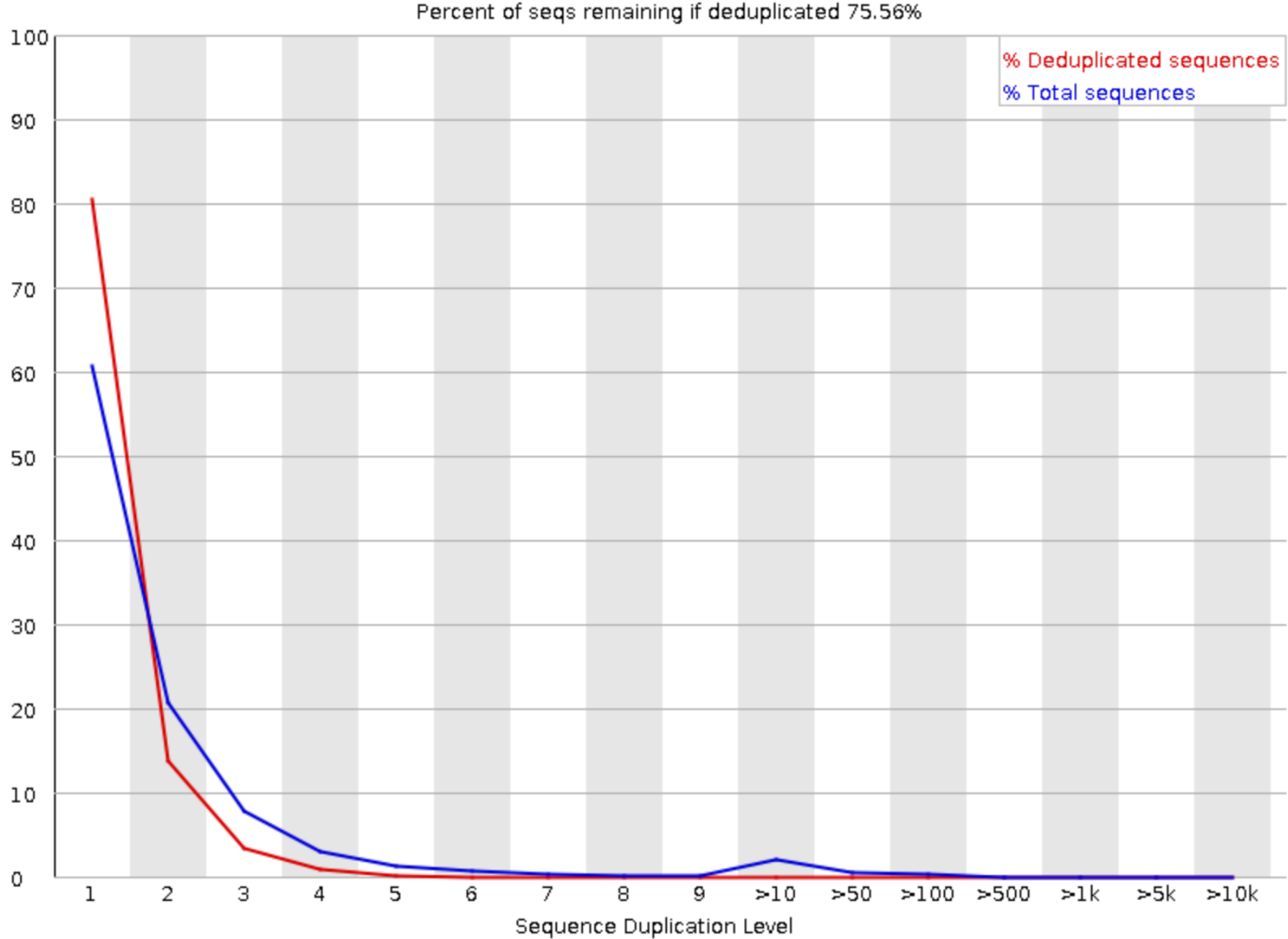
✔ **Per sequence GC content**

Summary

- ✔ [Basic Statistics](#)
- ✔ [Per base sequence quality](#)
- ✘ [Per tile sequence quality](#)
- ✔ [Per sequence quality scores](#)
- ✘ [Per base sequence content](#)
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- ✔ [Sequence Length Distribution](#)
- ✔ [Sequence Duplication Levels](#)
- ✔ [Overrepresented sequences](#)
- ✔ [Adapter Content](#)



✔ **Sequence Duplication Levels**



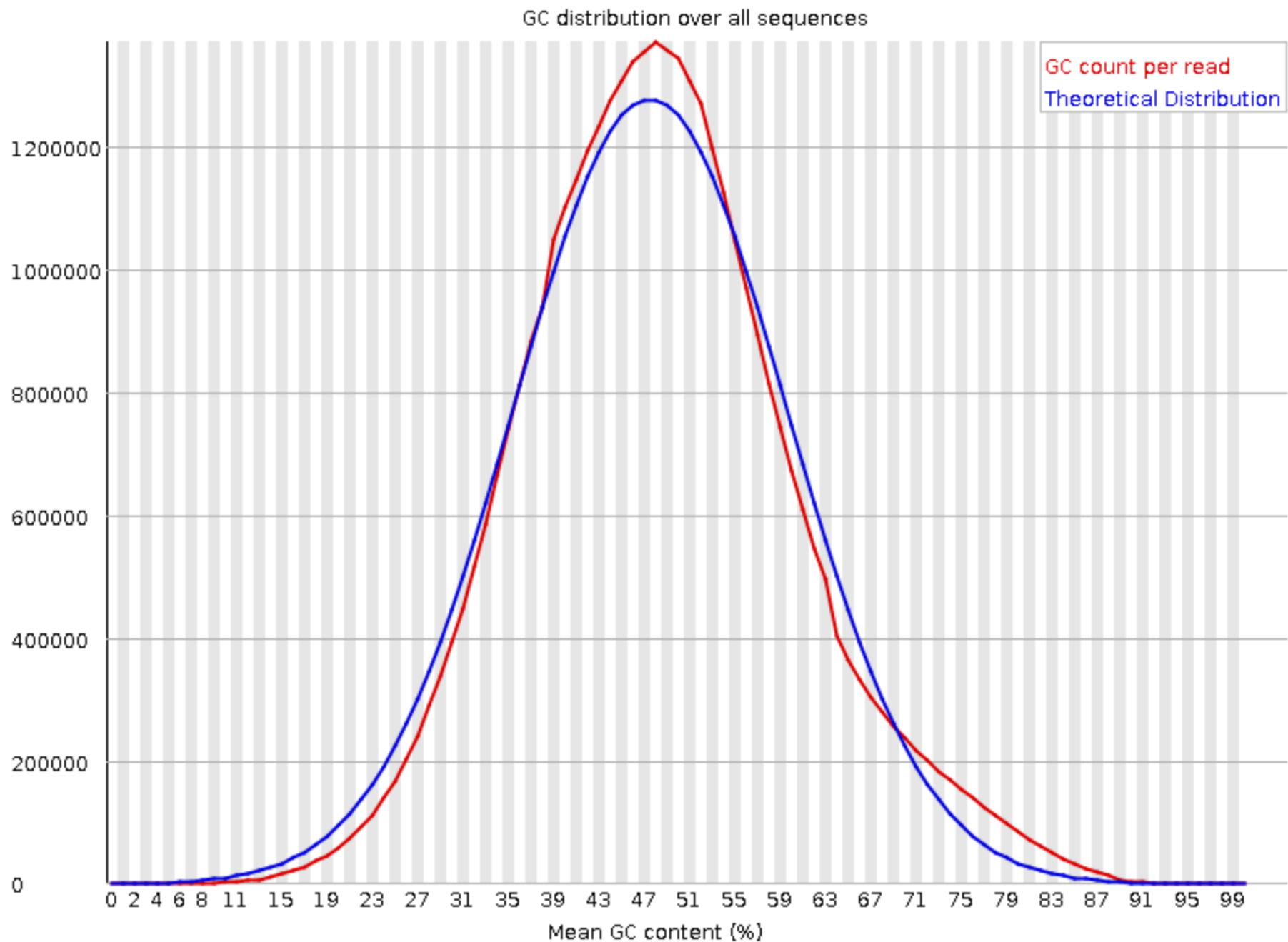
FastQC is used to asses the quality of the sequencing reads associated with the samples

GC distribution of data matches theoretical
GC distribution from genome (i.e., no bias)

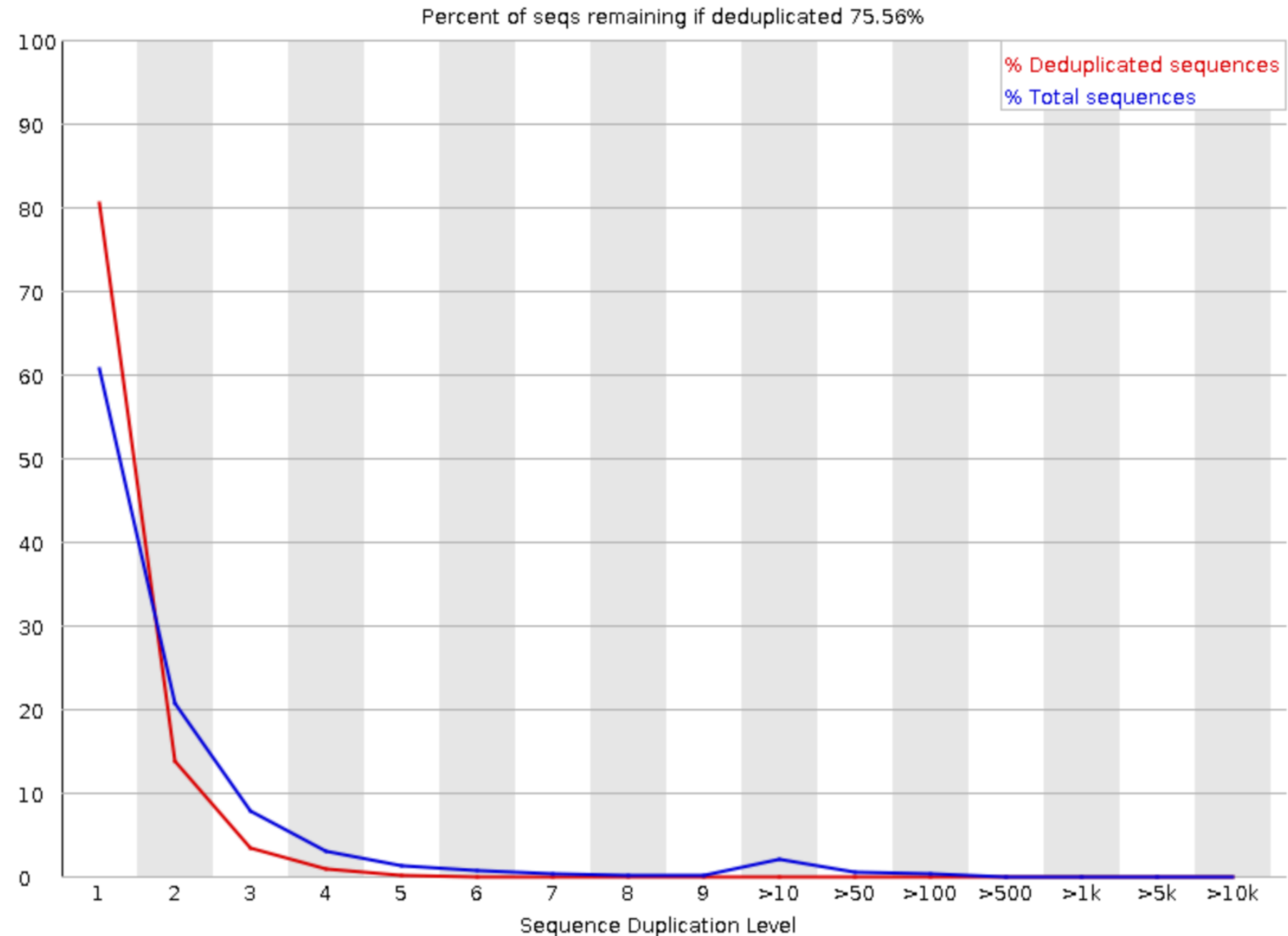
✔ **Per sequence GC content**

Summary

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- ✔ [Per base sequence quality](#)
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- ✔ [Adapter Content](#)



✔ **Sequence Duplication Levels**

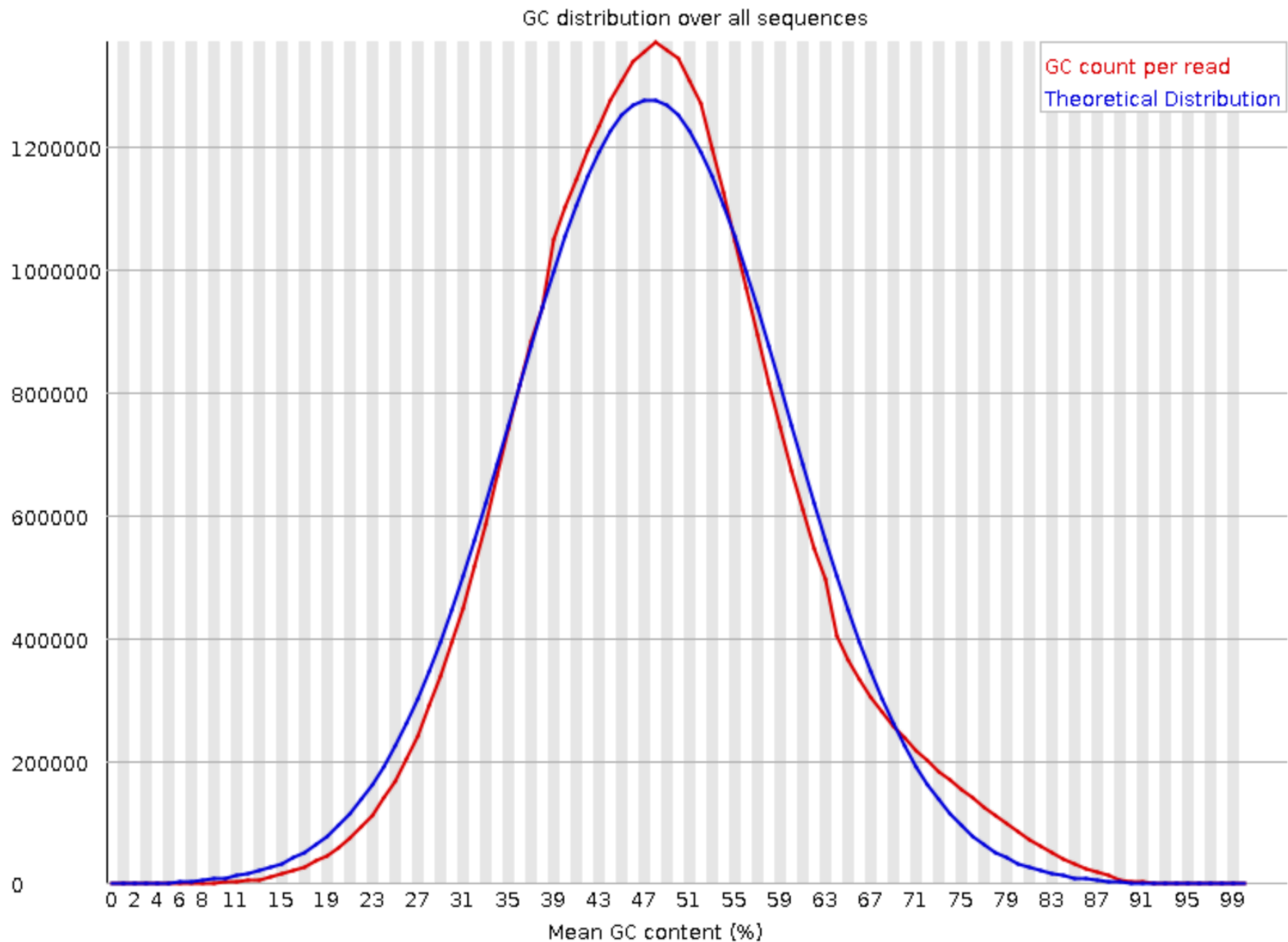


ATAC-seq data can sometimes be associated with GC bias so it is important to assess this metric in the FastQC report

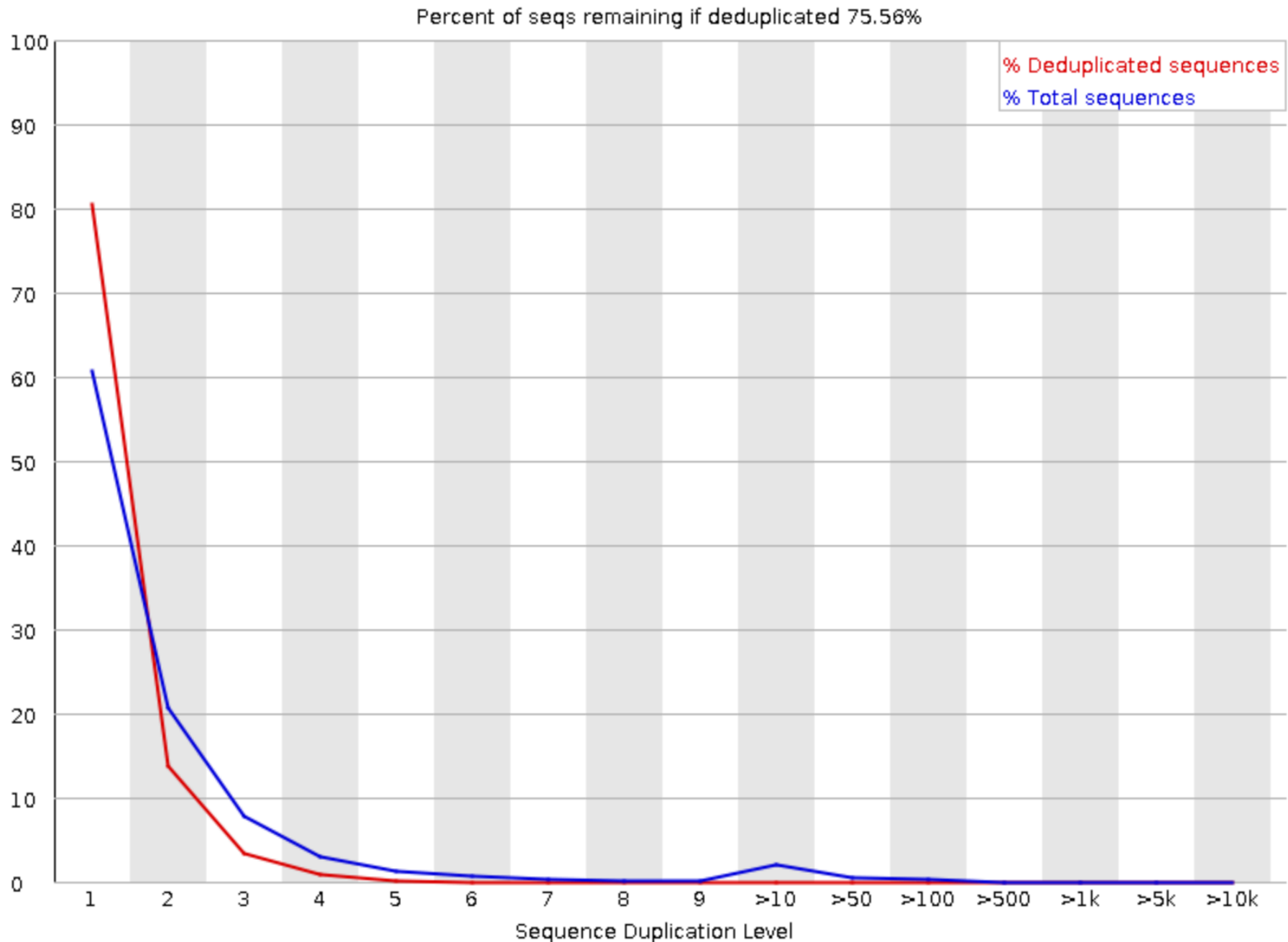
FastQC is used to asses the quality of the sequencing reads associated with the samples

GC distribution of data matches theoretical
GC distribution from genome (i.e., no bias)

✔ **Per sequence GC content**



✔ **Sequence Duplication Levels**



What do you think this would look like for a
DNA methyltransferase ChIP-seq?

FastQC is used to asses the quality of the sequencing reads associated with the samples

GC distribution of data matches theoretical
GC distribution from genome (i.e., no bias)

✔ **Per sequence GC content**

Summary

✔

[Basic Statistics](#)

✔

[Per base sequence quality](#)

✖

[Per tile sequence quality](#)

✔

[Per sequence quality scores](#)

✖

[Per base sequence content](#)

✔

[Per sequence GC content](#)

✔

[Per base N content](#)

✔

[Sequence Length Distribution](#)

✔

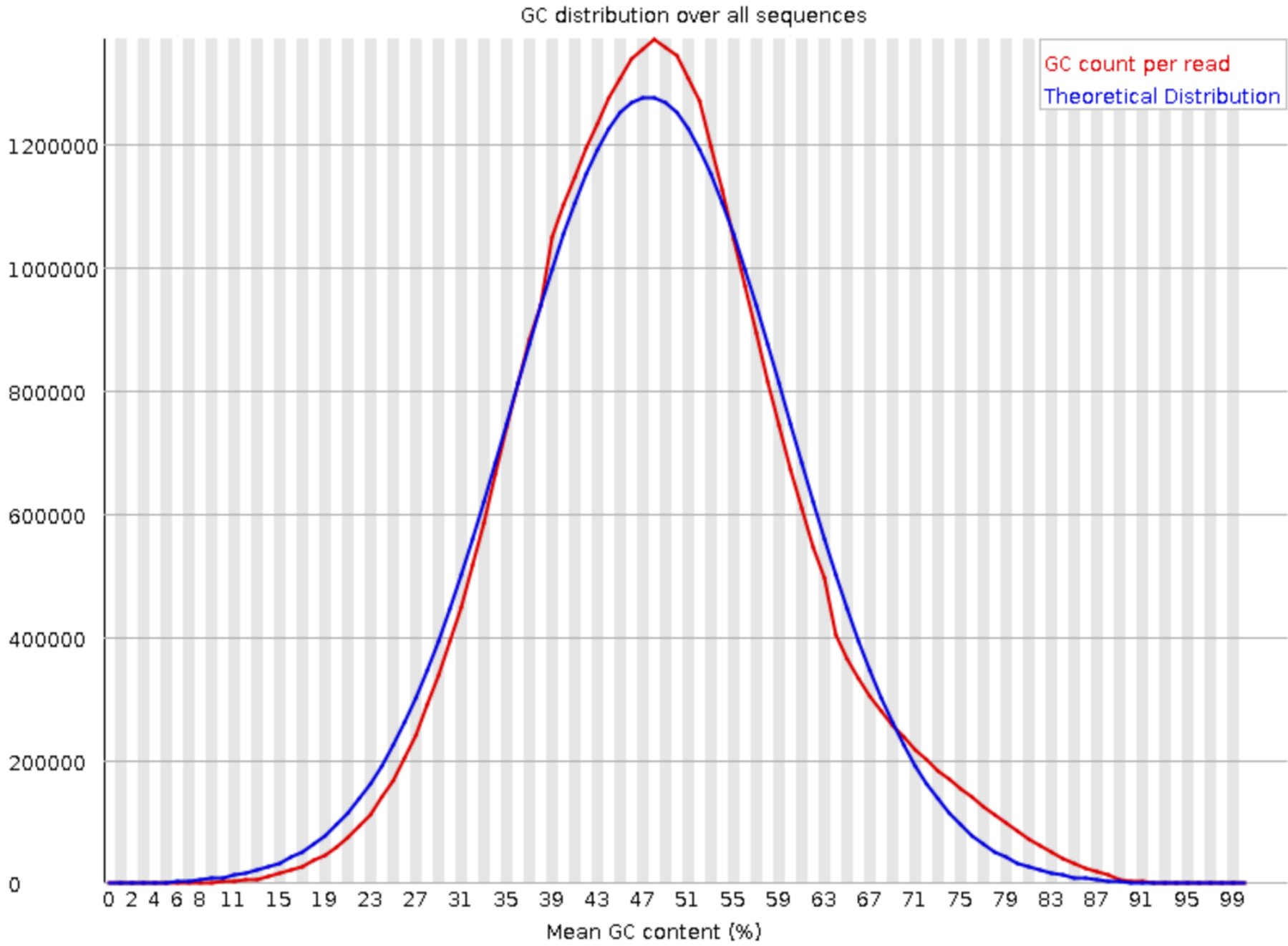
[Sequence Duplication Levels](#)

✔

[Overrepresented sequences](#)

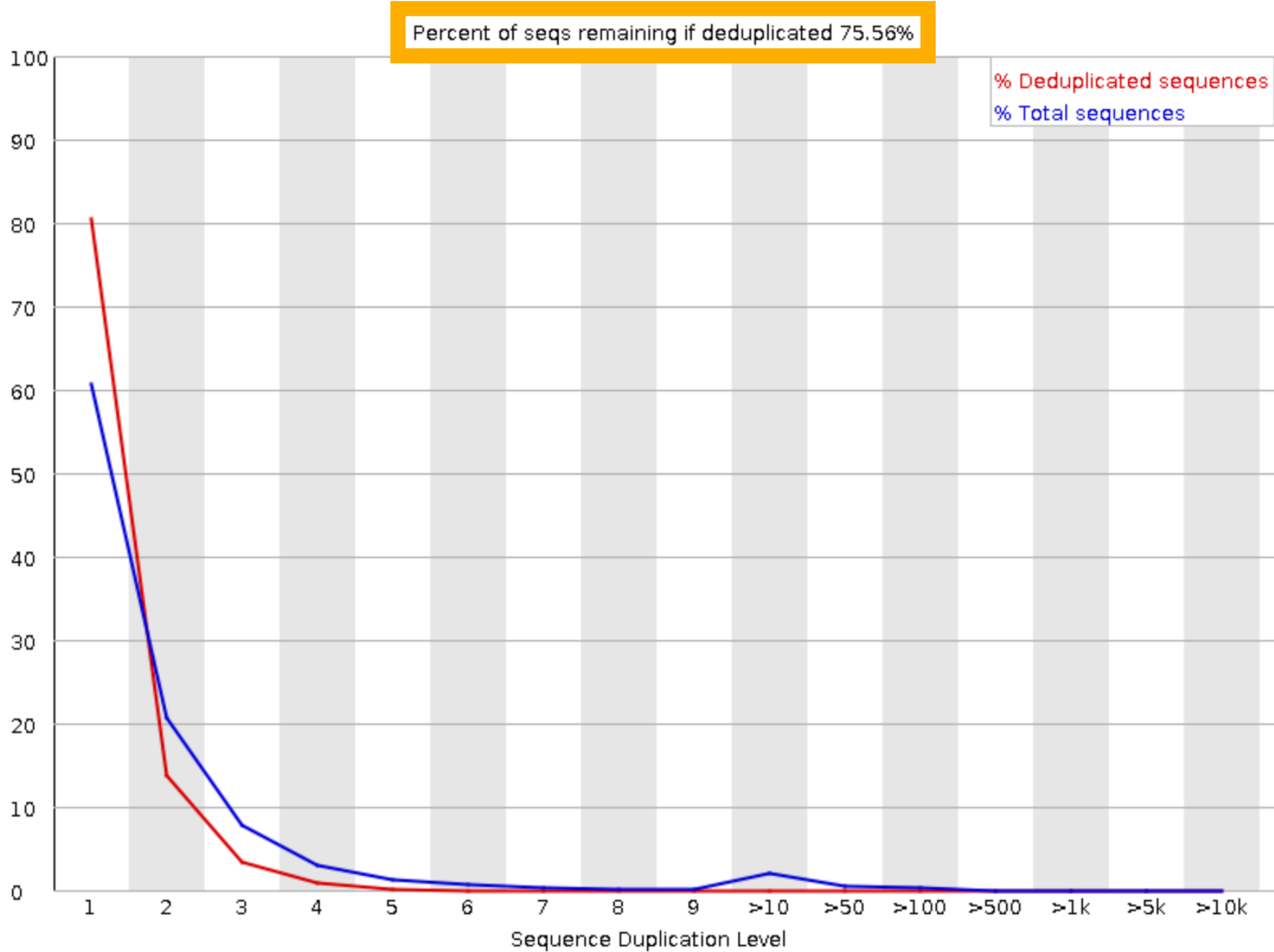
✔

[Adapter Content](#)



~75% of sequences remaining if de-duplicated

✔ **Sequence Duplication Levels**



What do you think this would look like for a
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FastQC is used to asses the quality of the sequencing reads associated with the samples

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✔ **Per sequence GC content**

Summary

✔

[Basic Statistics](#)

✔

[Per base sequence quality](#)

✖

[Per tile sequence quality](#)

✔

[Per sequence quality scores](#)

✖

[Per base sequence content](#)

✔

[Per sequence GC content](#)

✔

[Per base N content](#)

✔

[Sequence Length Distribution](#)

✔

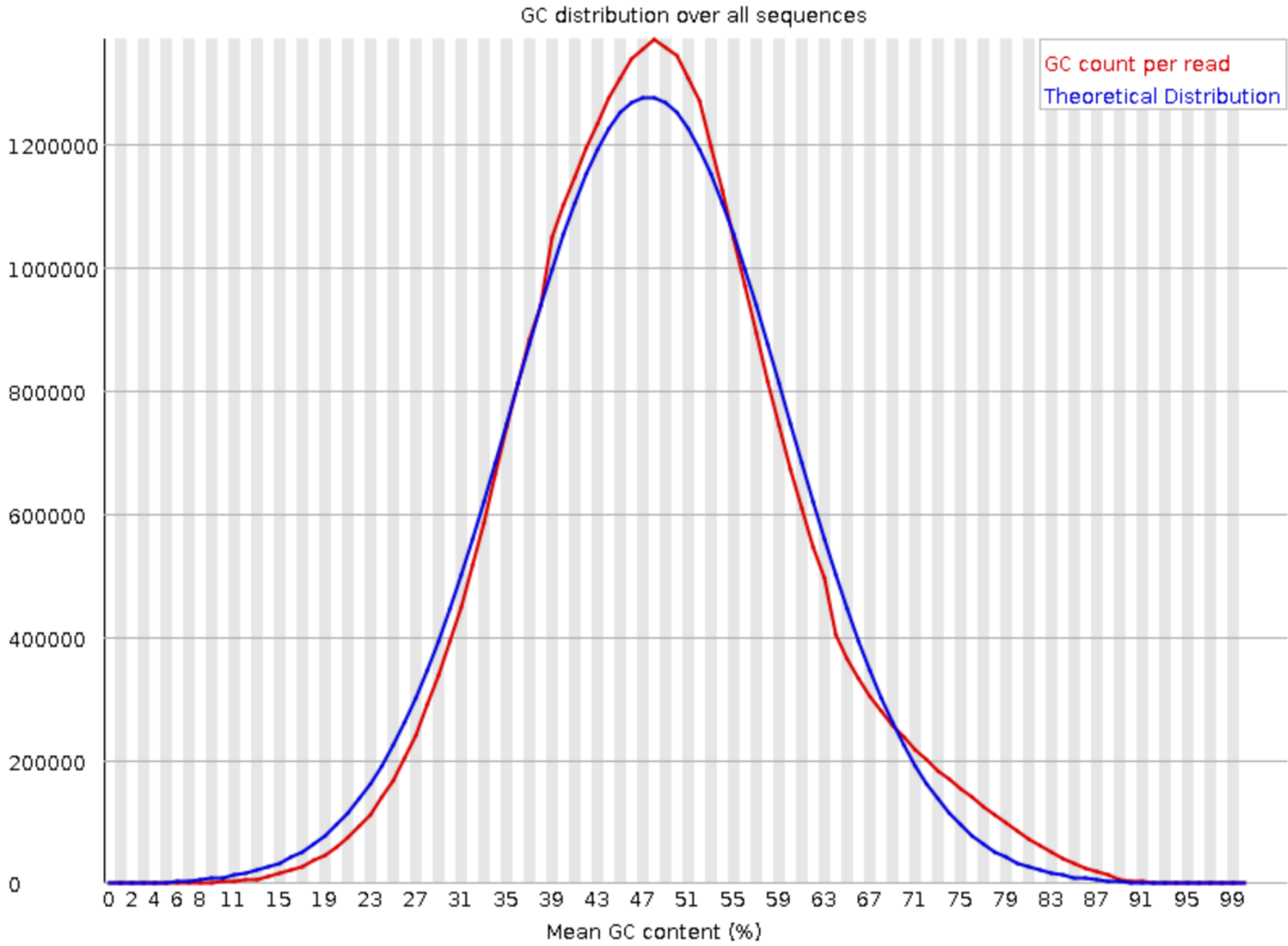
[Sequence Duplication Levels](#)

✔

[Overrepresented sequences](#)

✔

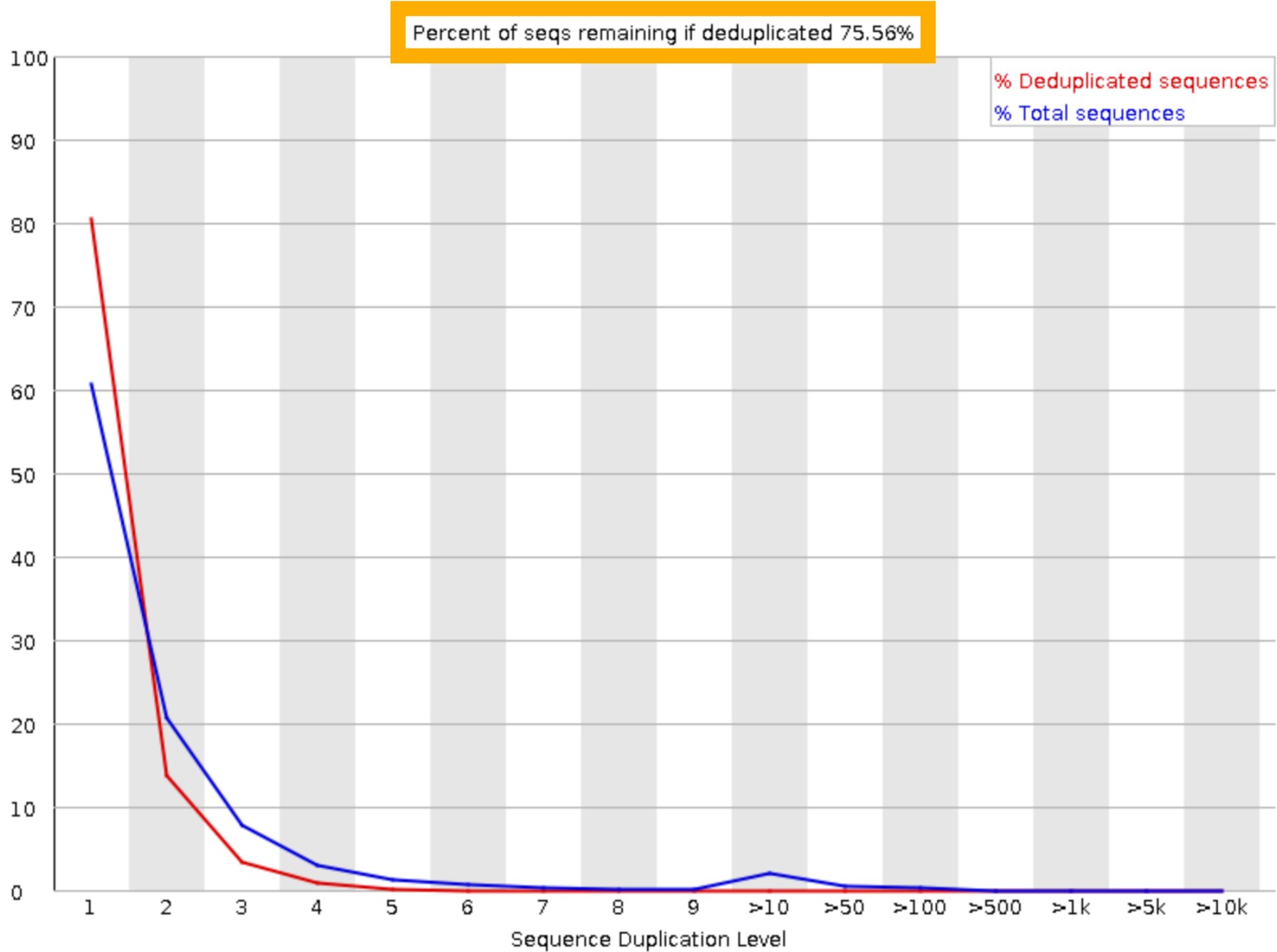
[Adapter Content](#)



What do you think this would look like for a
DNA methyltransferase ChIP-seq?

~75% of sequences remaining if de-duplicated

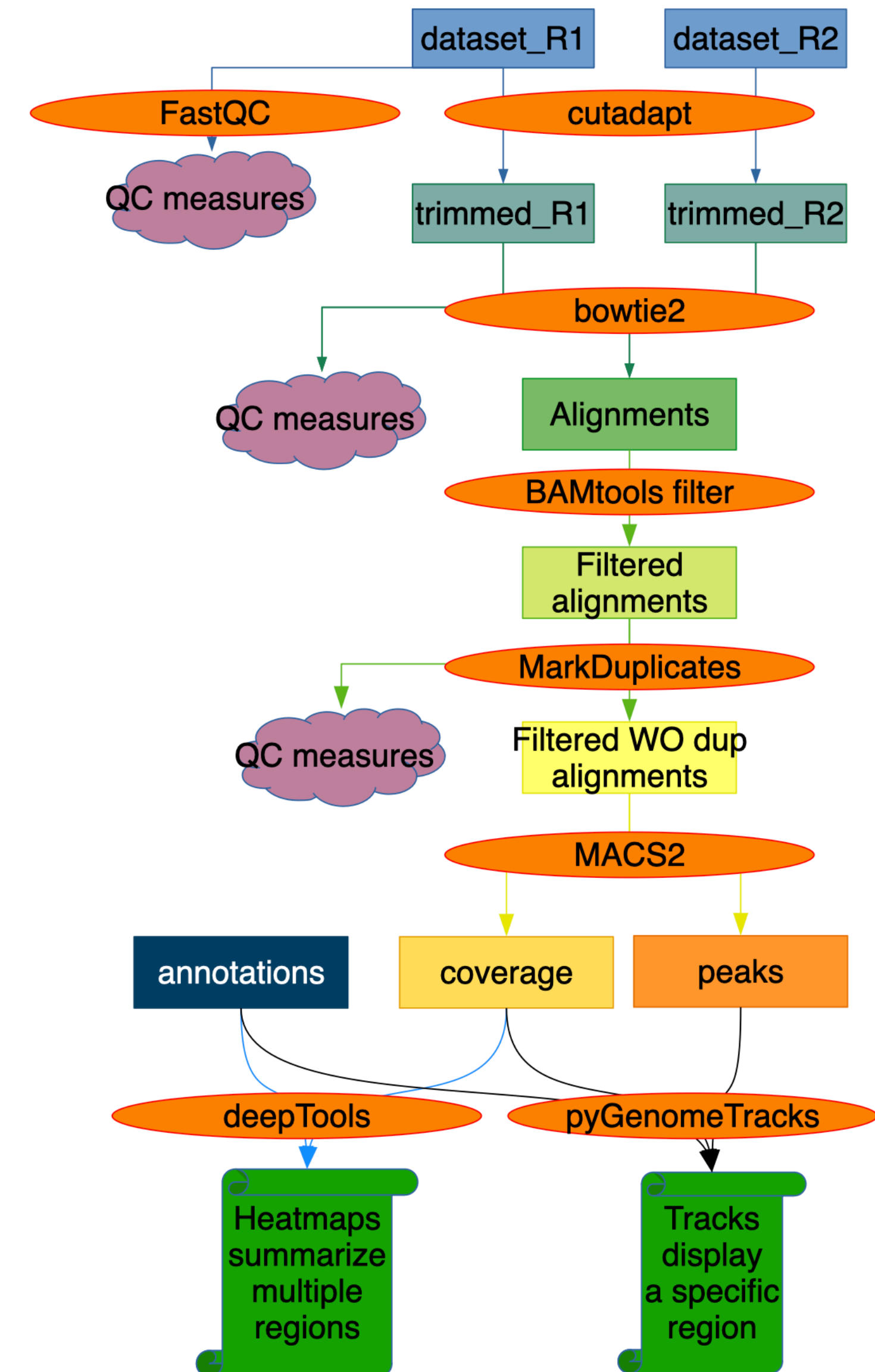
✔ **Sequence Duplication Levels**



What do you think this would look like for a
RNA-seq data?

ATAC-seq data processing script file/commands

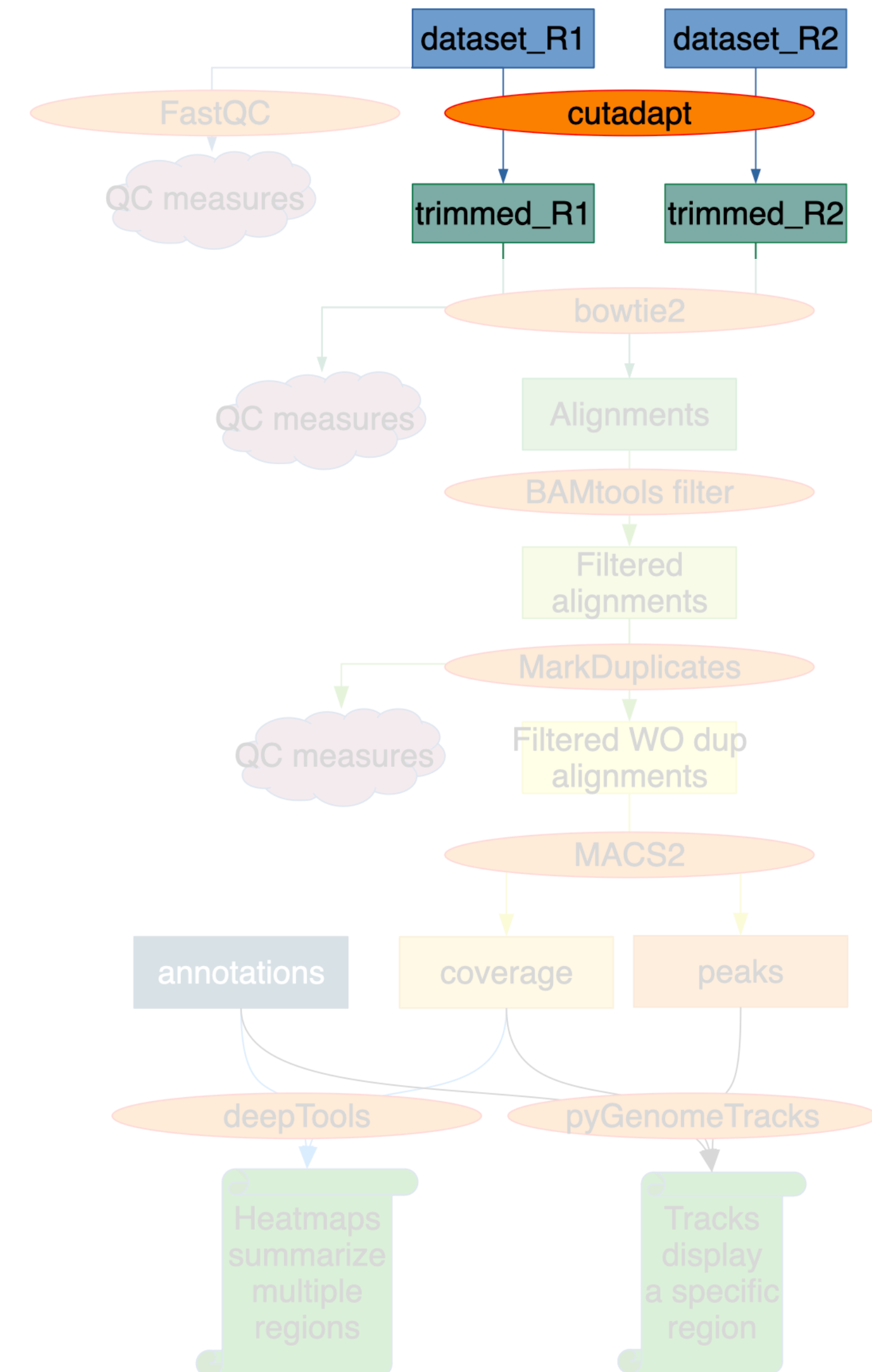
Heyl and Delisle, Galaxy Training



ATAC-seq data processing script file/commands

```
cutadapt \  
  -a AGATCGGAAGAGCACACGTCTGAACTCCAGTCA \  
  -A AGATCGGAAGAGCGTCGTGTAGGGAAAGAGTGT \  
  --minimum-length=25 -j [number of threads] \  
  -o [output dir for read 1] -p [output dir for read 2] \  
    [input read 1 fastq] [input read 2 fastq] \  
> [output for cut adapt report]
```

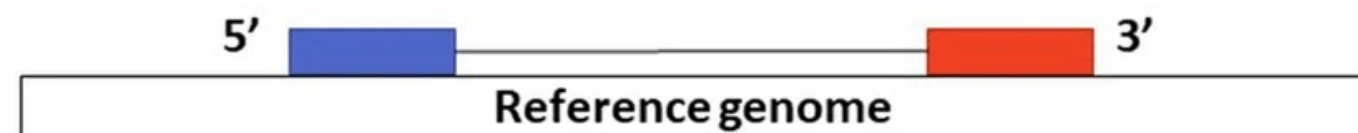
Heyl and Delisle, Galaxy Training



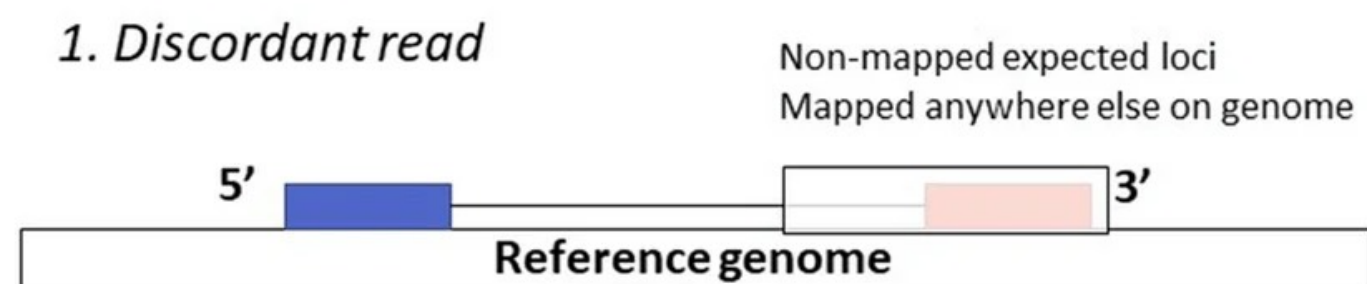
ATAC-seq data processing script file/commands

```
bowtie2 --local --very-sensitive-local \
--no-unal --no-mixed --no-discordant \
-x [input for genome index] -I 10 -X 1000 \
-1 [input trimmed read 1] -2 [input trimmed read 2] \
--threads [number of threads] \
| samtools view -@ [number of threads] -F 780 -f 2 -bh - \
| samtools sort -@ [number of threads] -T [output directory for BAM file]
```

Proper pairs



Non-proper pairs



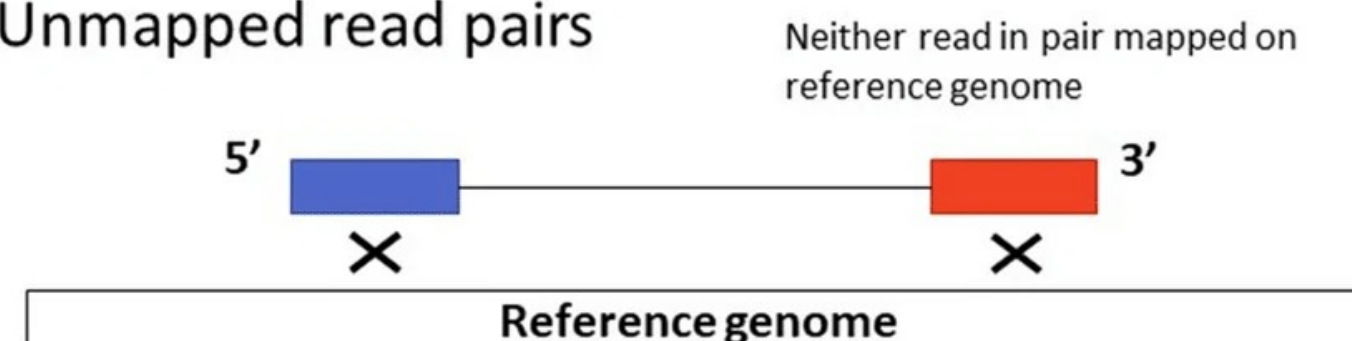
2. Split read



Singleton

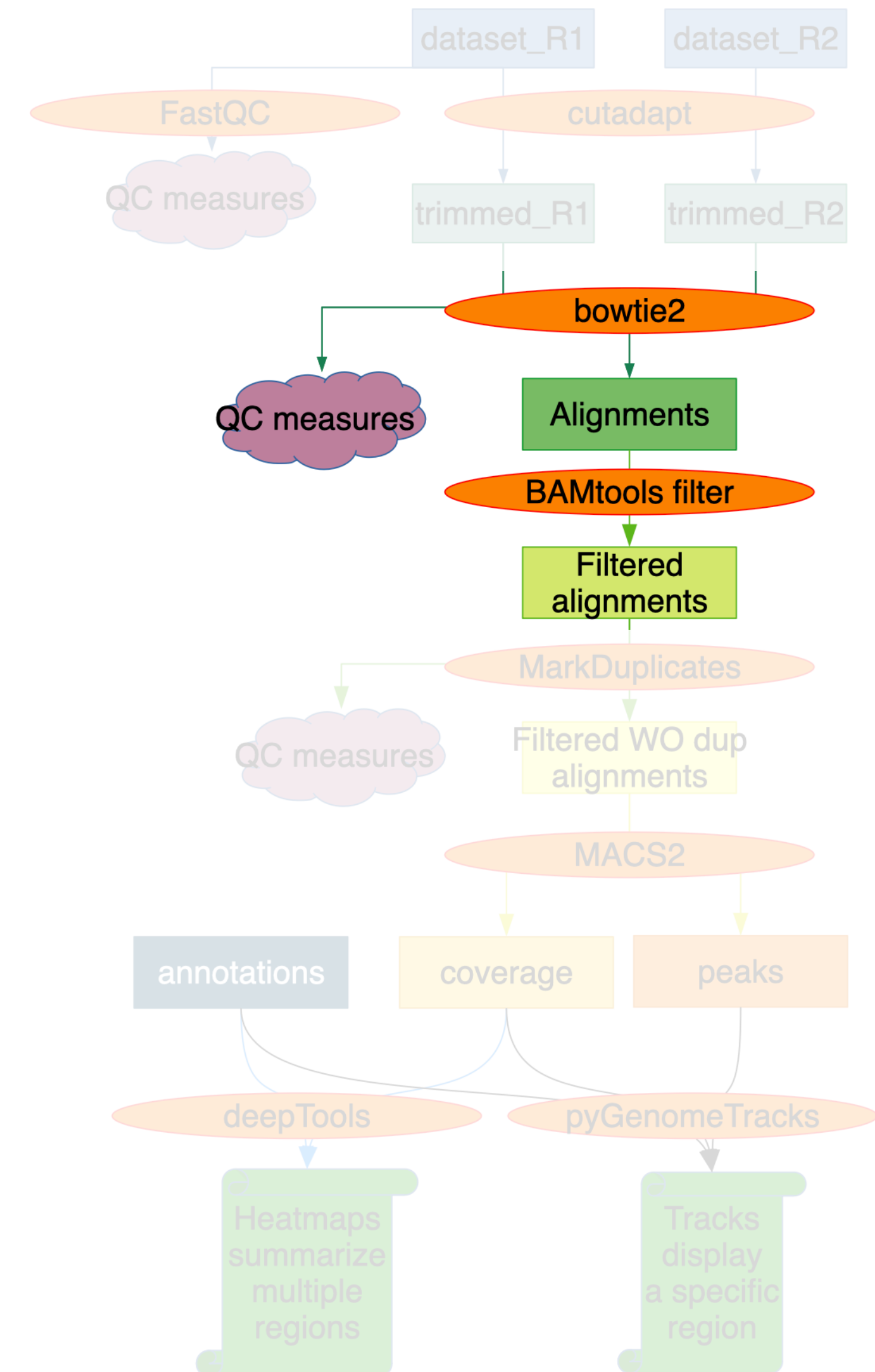


Unmapped read pairs



Ishihara et al., 2022

Heyl and Delisle, Galaxy Training

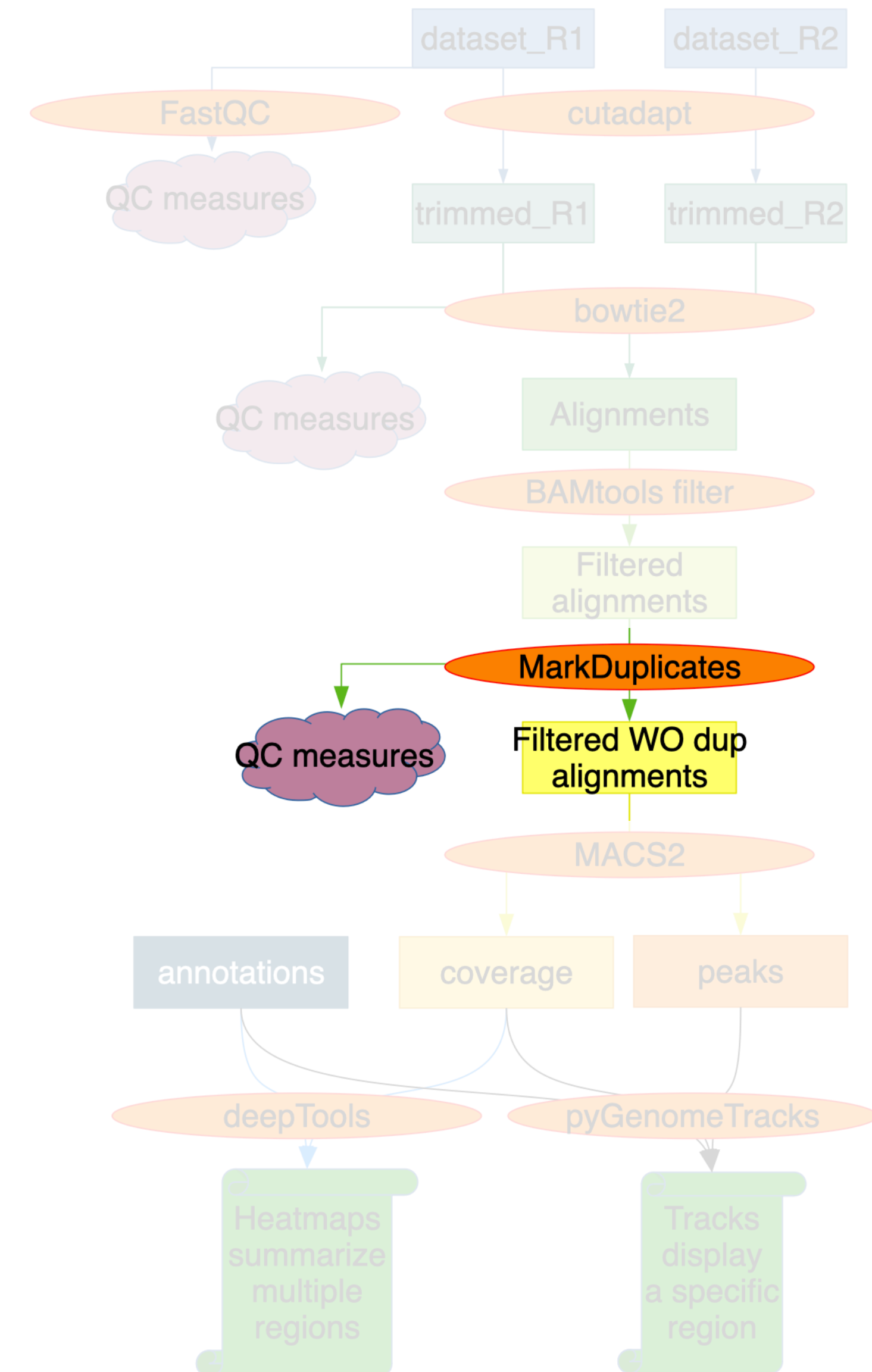


ATAC-seq data processing script file/commands

Heyl and Delisle, Galaxy Training

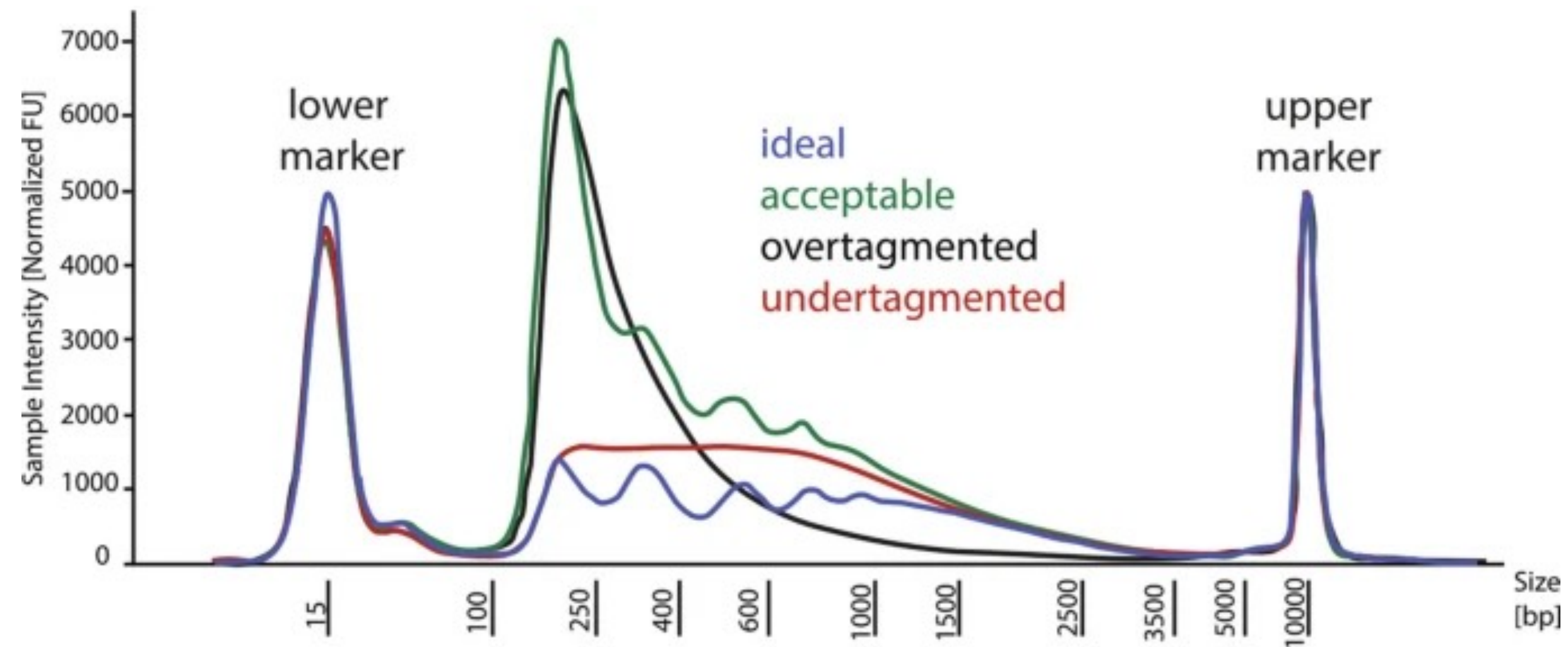
```
picard MarkDuplicates \
  I=[input BAM file] \
  O=[output dir for BAM file with marked duplicate reads] \
  M=[output dir for stats file] \
  VALIDATION_STRINGENCY=SILENT

samtools view -@ [number of threads] -F 1804 -f 2 -b [input BAM file] \
| samtools sort -@ [number of threads] -o [output no duplicate BAM file] \
samtools index [output indexed no duplicate BAM file]
```



Nucleosomal laddering is an important QC metric of good quality ATAC-seq data

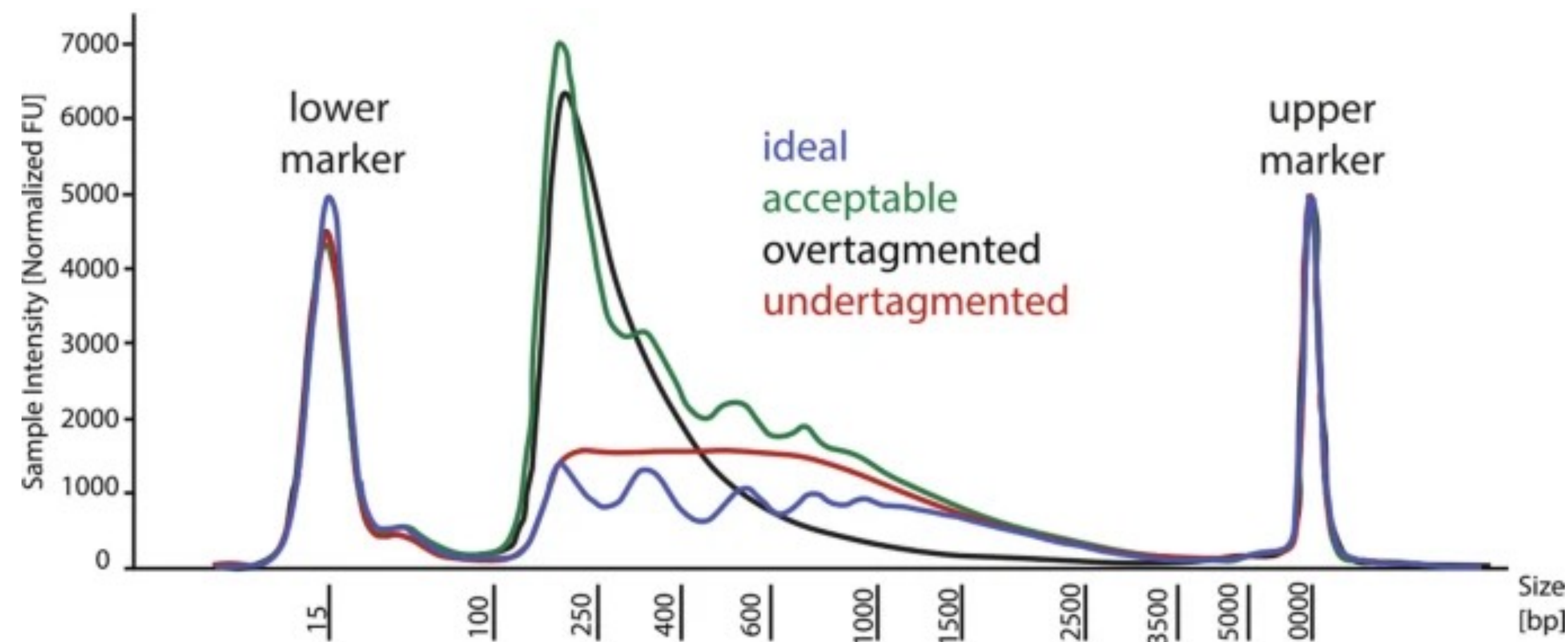
Fragment size distribution of amplified ATAC-seq library



Gehrke and Srivastava, 2022

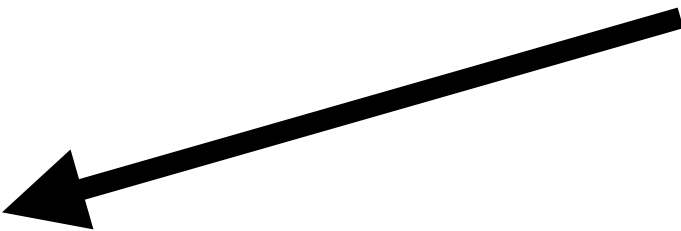
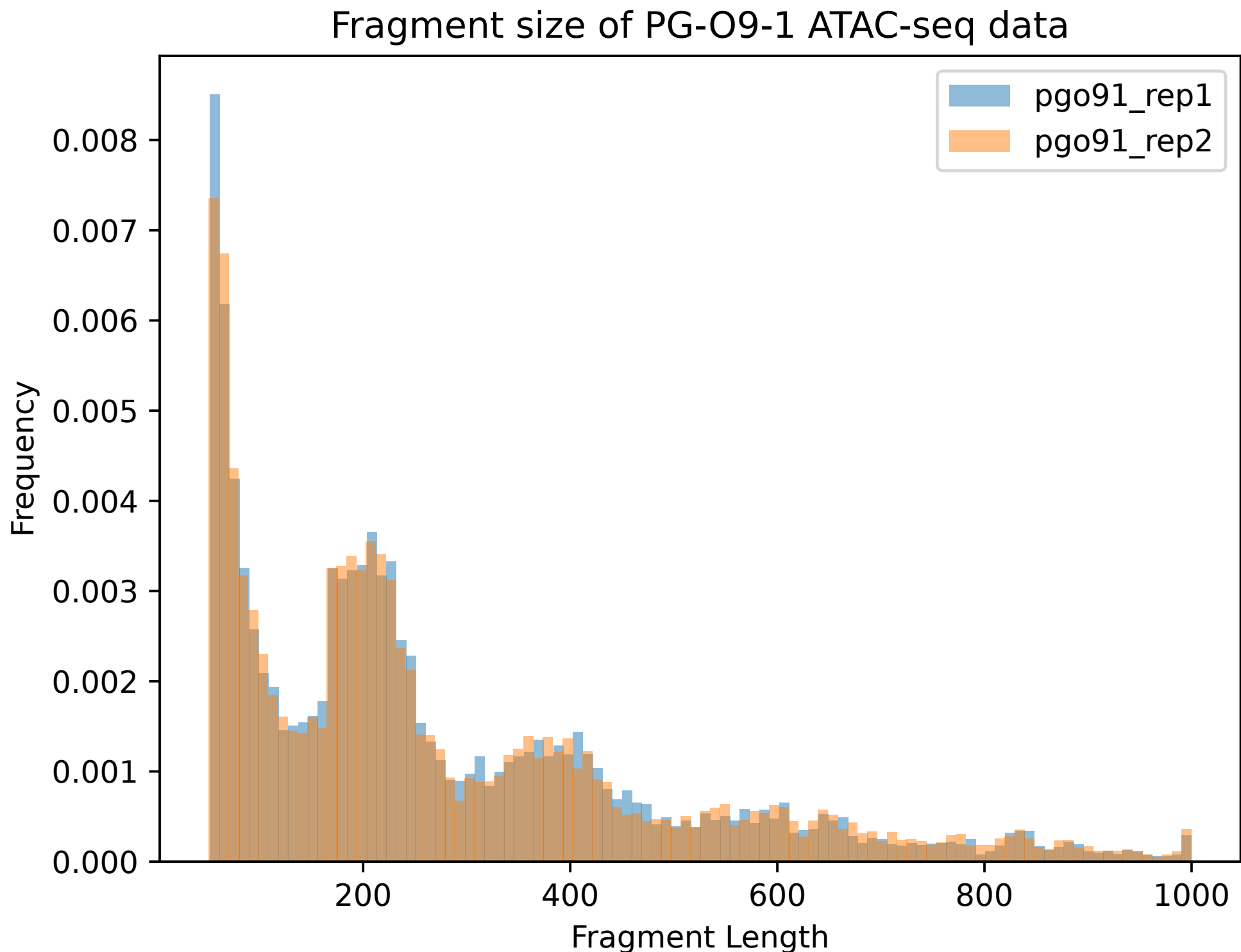
Nucleosomal laddering is an important QC metric of good quality ATAC-seq data

Fragment size distribution of amplified ATAC-seq library



Gehrke and Srivastava, 2022

Fragment size distribution of sequencing data
(from the data we will analyze today)



```
bamPEFragmentSize -b sample1.bam sample2.bam -o hist.png
```

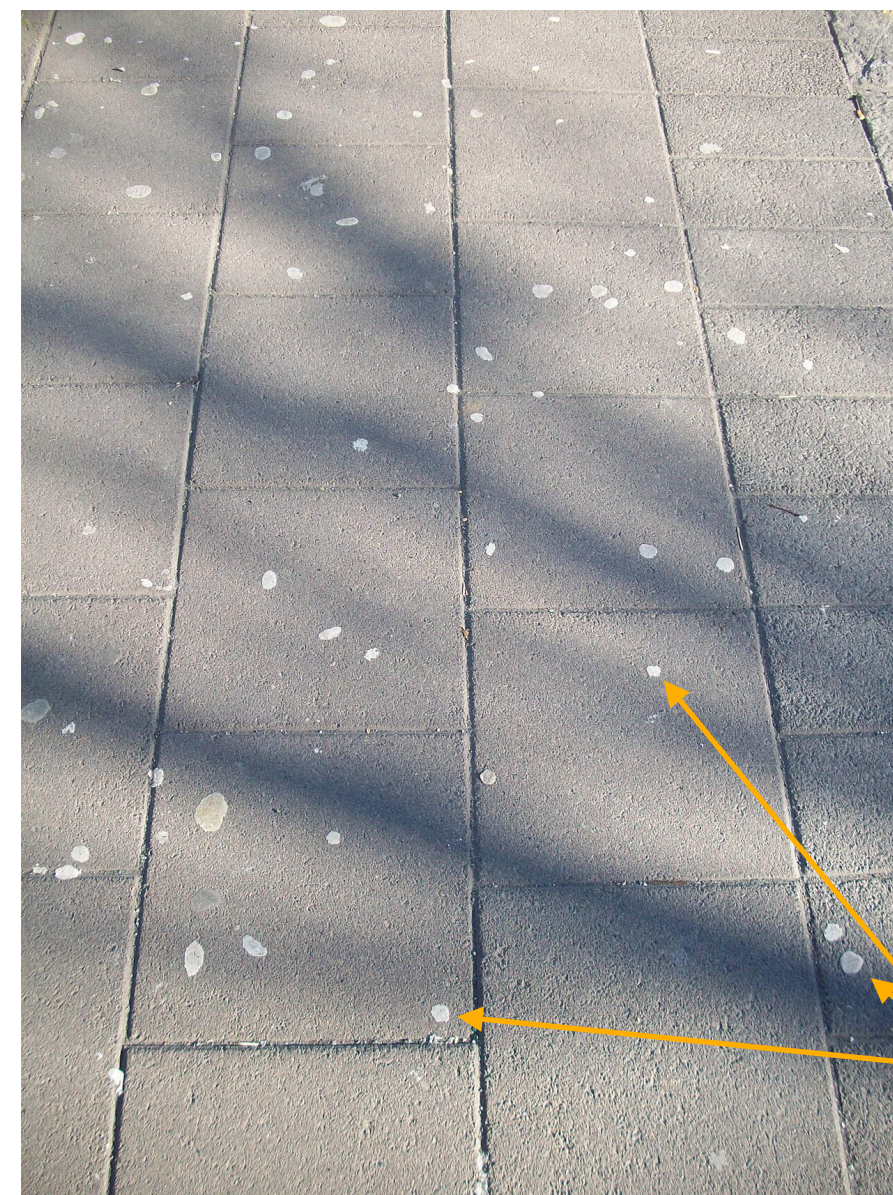
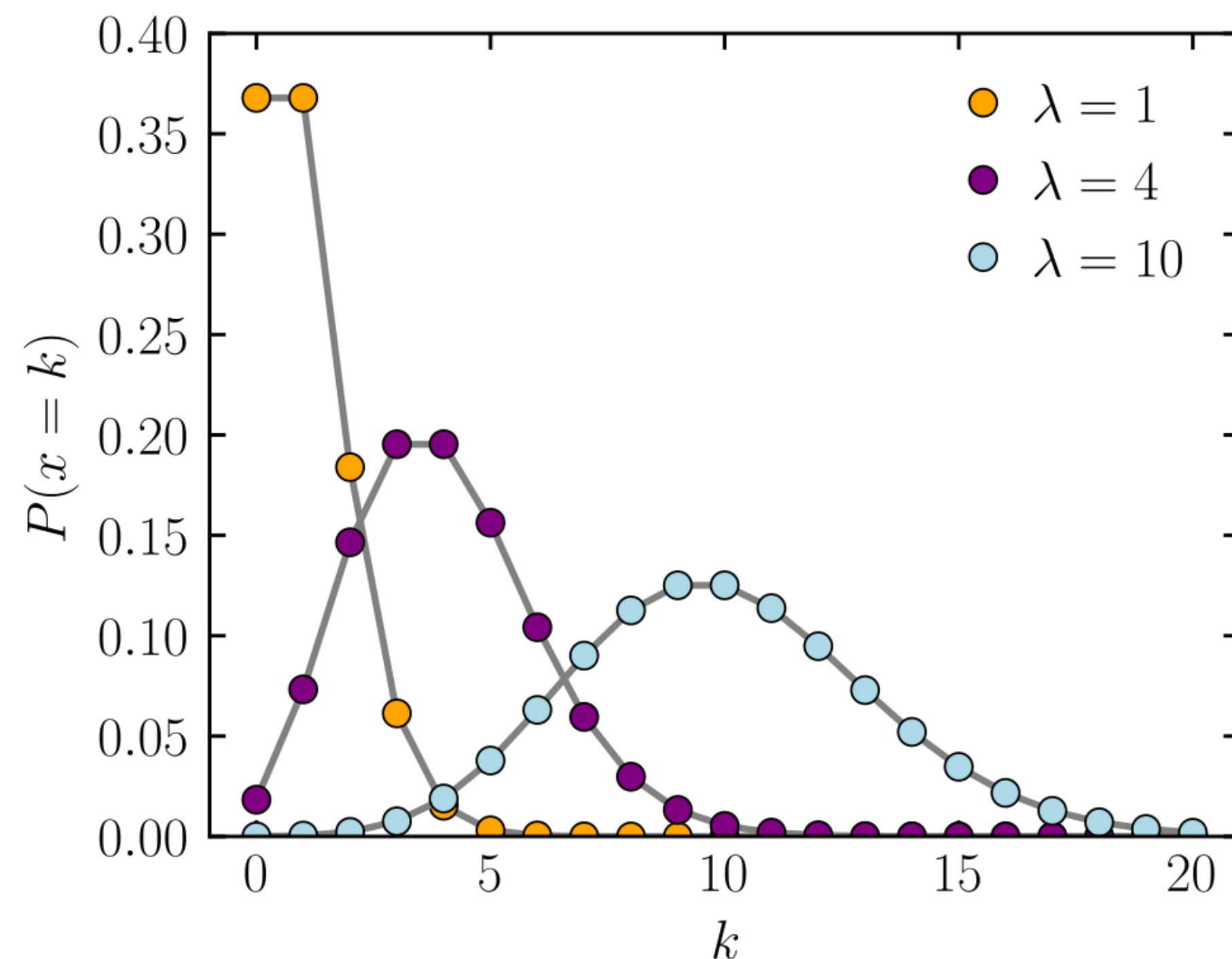

ATAC-seq data processing script file/commands

p-value is calculated by measuring the difference between
observed data and what you **expect** to see by random **chance**

$$P_{\lambda}(X=k) = \frac{\lambda^k}{k! * e^{-\lambda}} \quad (\text{Poisson distribution})$$

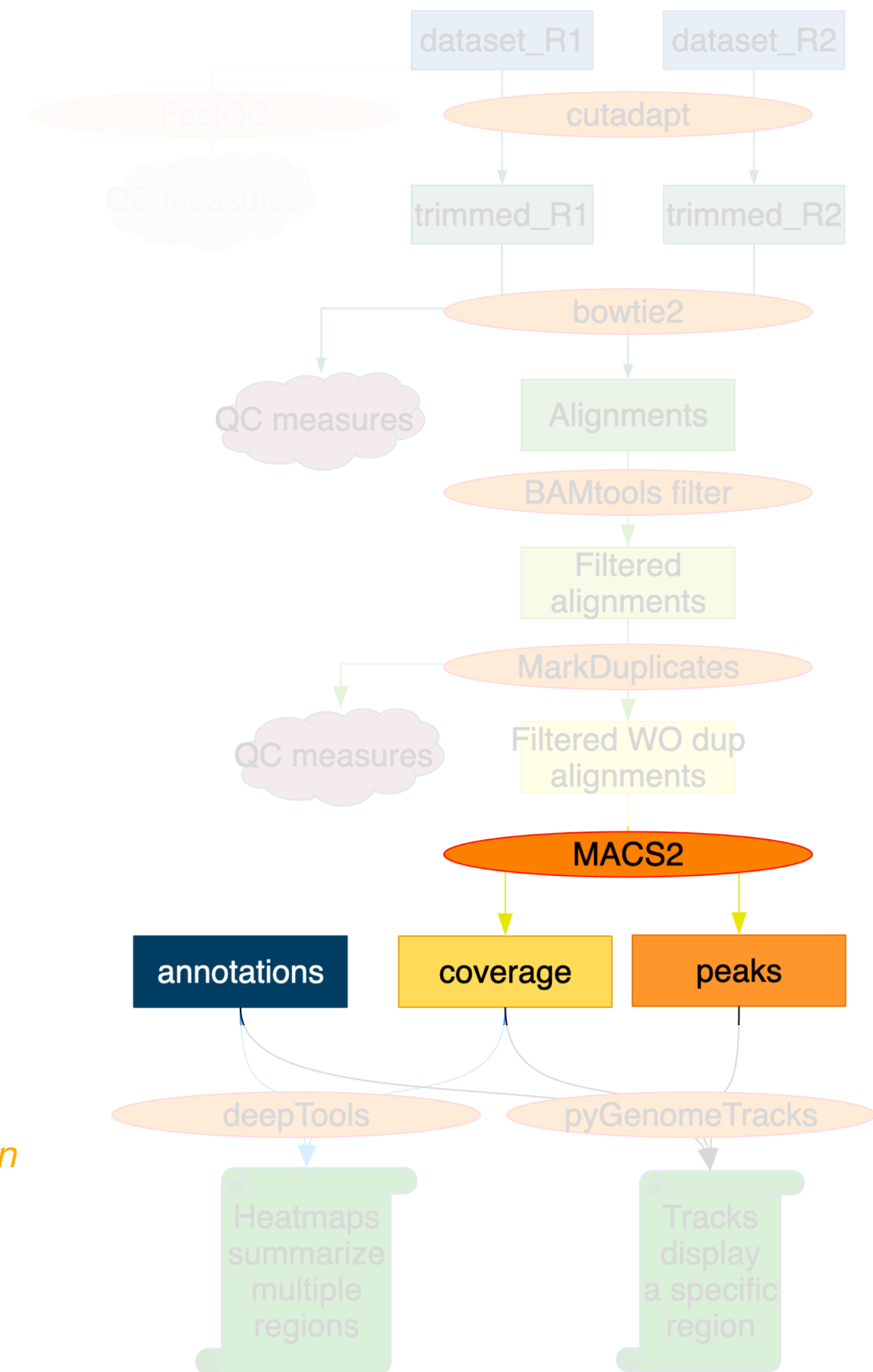
k = number of occurrences

λ = the expected rate of occurrences (number of reads in the genomic window)



Chewing gum on
tiles is Poisson
distributed

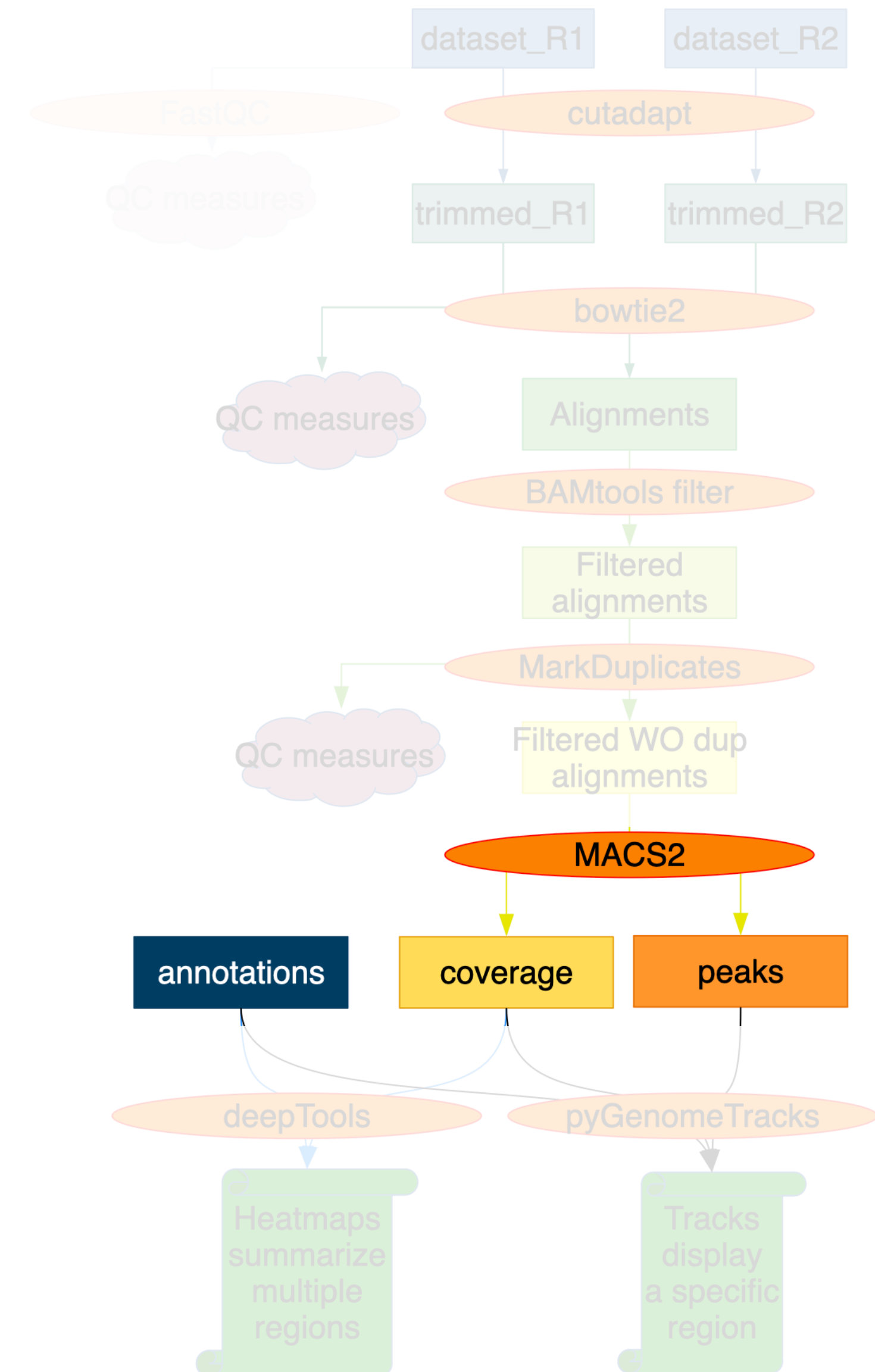
Heyl and Delisle, Galaxy Training



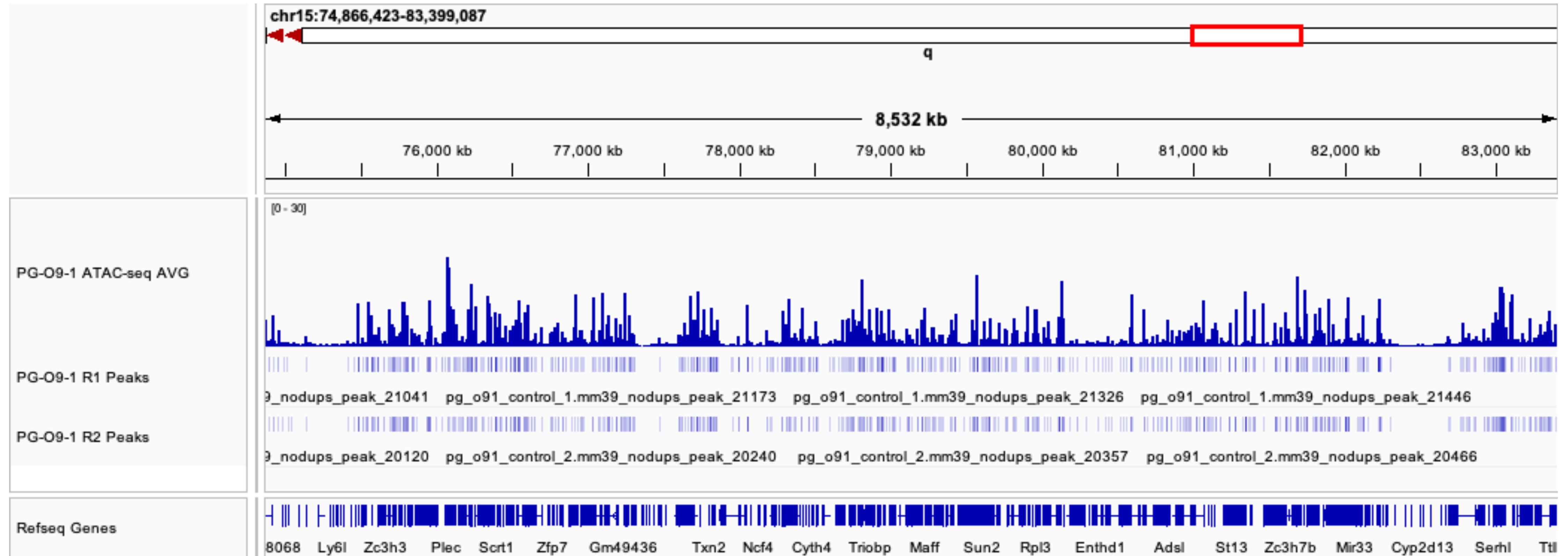
ATAC-seq data processing script file/commands

Heyl and Delisle, Galaxy Training

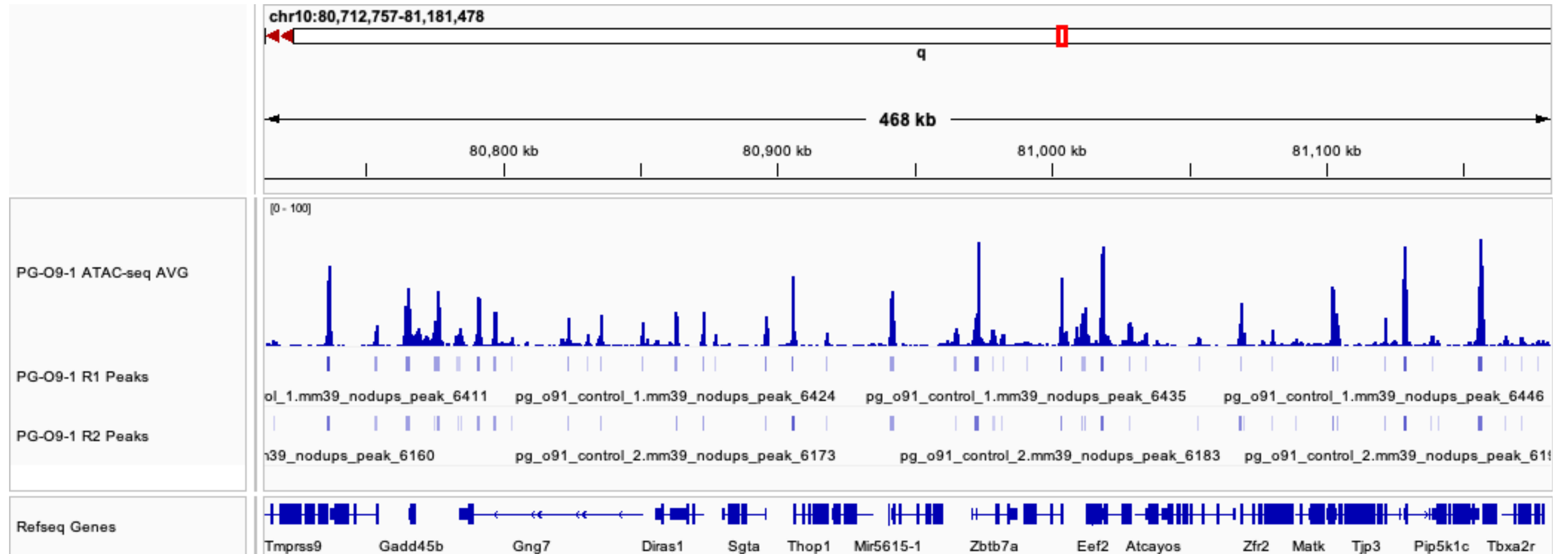
```
macs2 callpeak -t [input BAM file] \  
  -n [prefix for output files] \  
  -f BAMPE -g [effective genome size] \  
  -q [q-value for peak calling - 0.05] \  
  --call-summits \  
  --outdir [output directory]  
  
bamCoverage --bam [input BAM file] \  
  --outFileName [output dir for bigwig file] \  
  --outFileFormat [format of output file - bigwig] \  
  --binSize [size of genomic bin - 5kb] \  
  --numberOfProcessors [number of threads] \  
  --normalizeUsing [sequencing depth normalization approach - RPGC] \  
  --effectiveGenomeSize [genome size] \  
  --extendReads
```



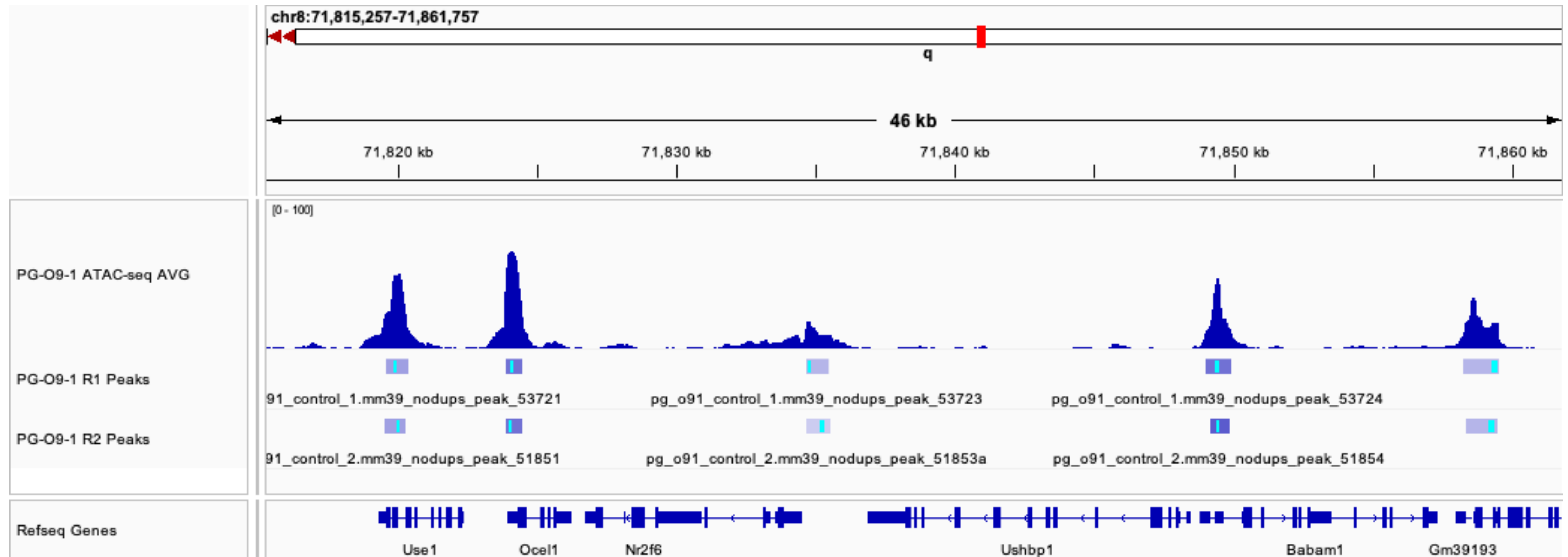
Assessing tracks and peaks in the genome browser



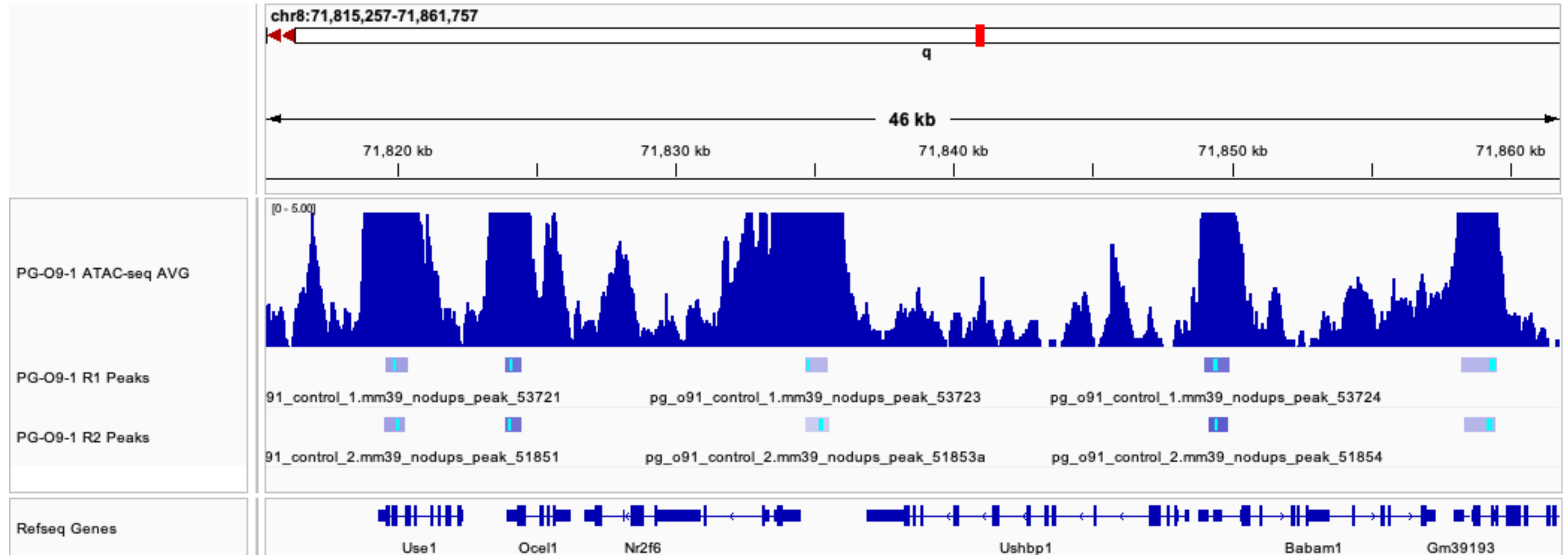
Assessing tracks and peaks in the genome browser



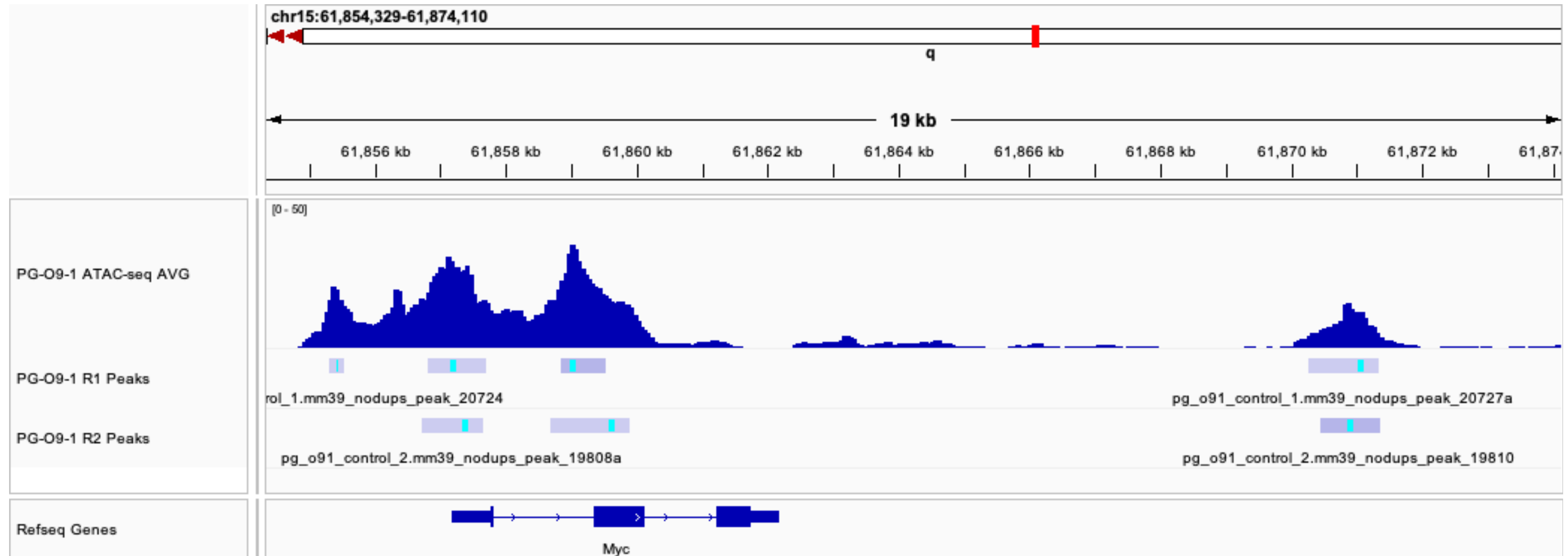
Assessing tracks and peaks in the genome browser



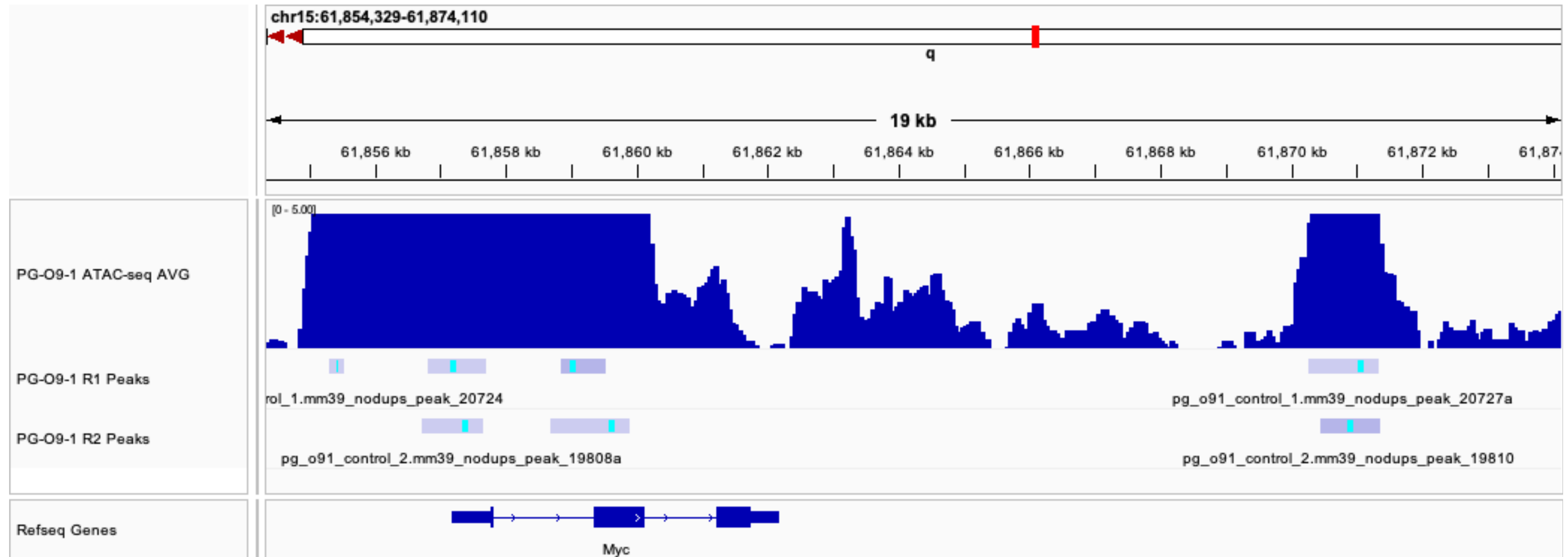
Assessing tracks and peaks in the genome browser



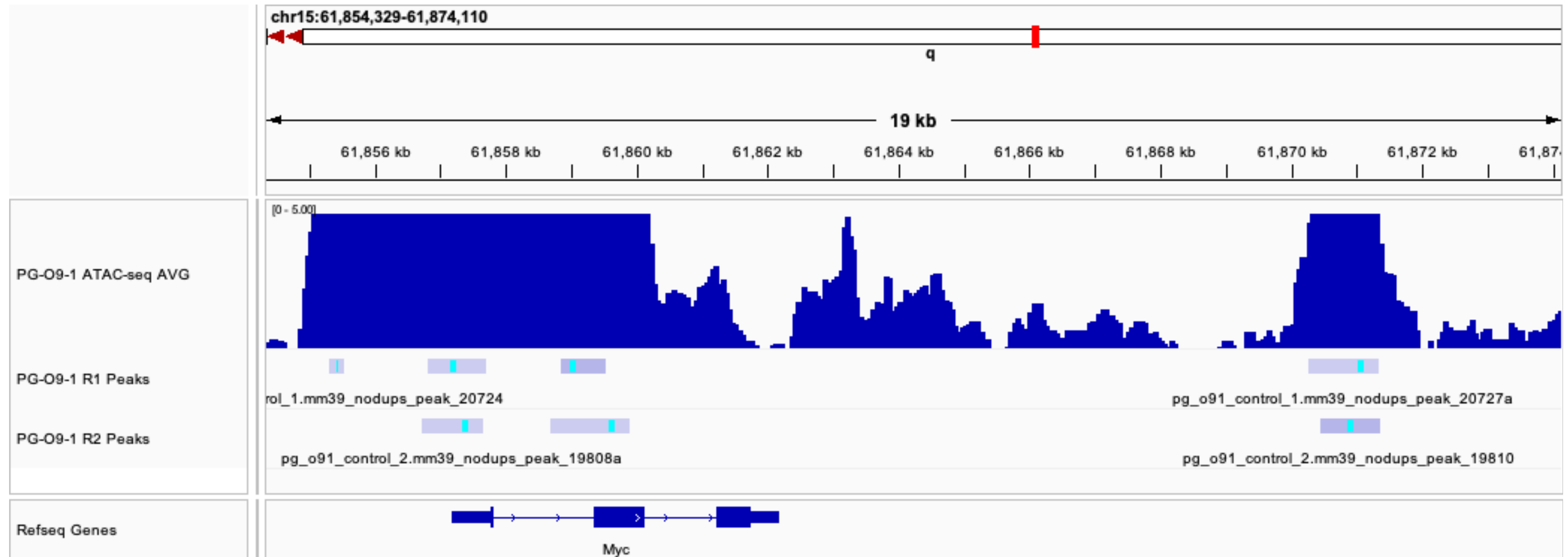
Assessing tracks and peaks in the genome browser



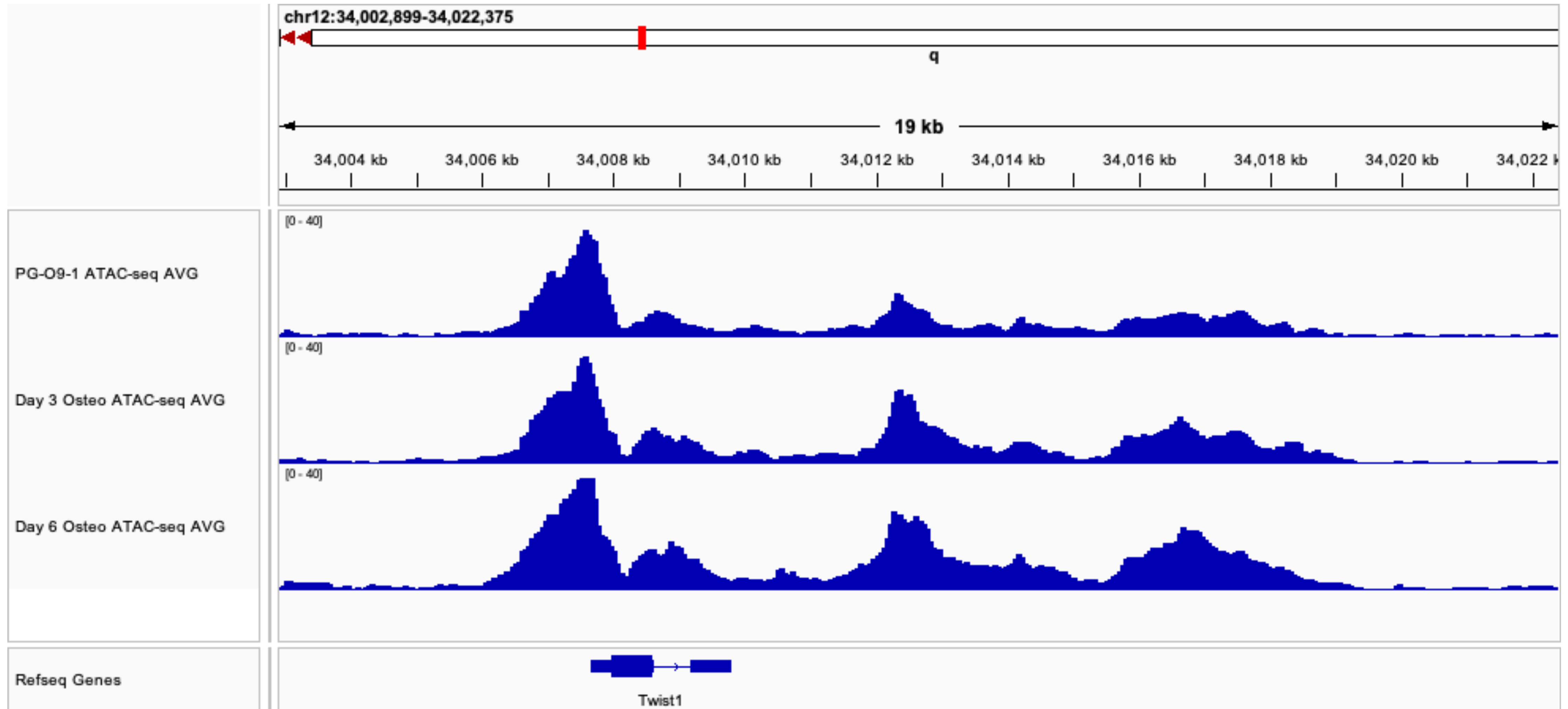
Assessing tracks and peaks in the genome browser



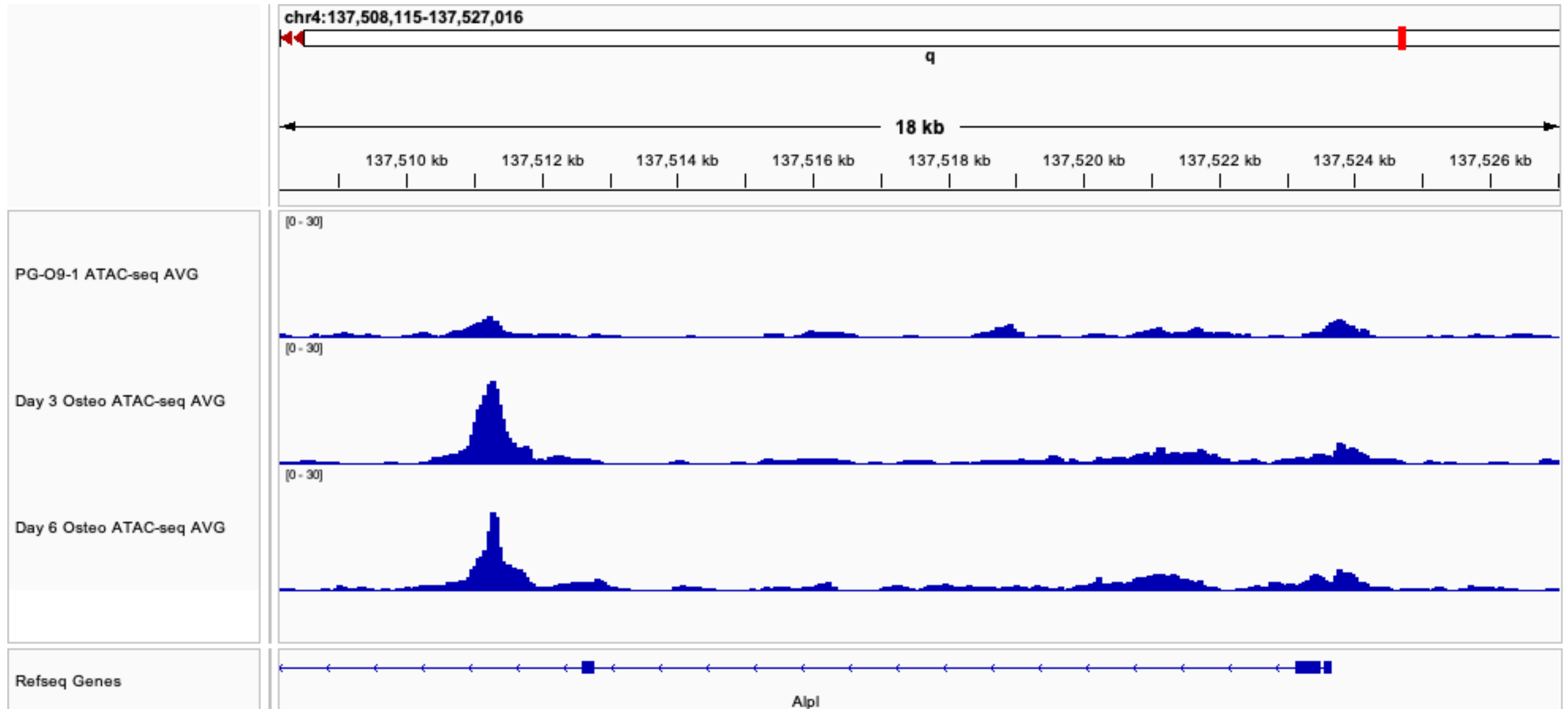
Assessing tracks and peaks in the genome browser



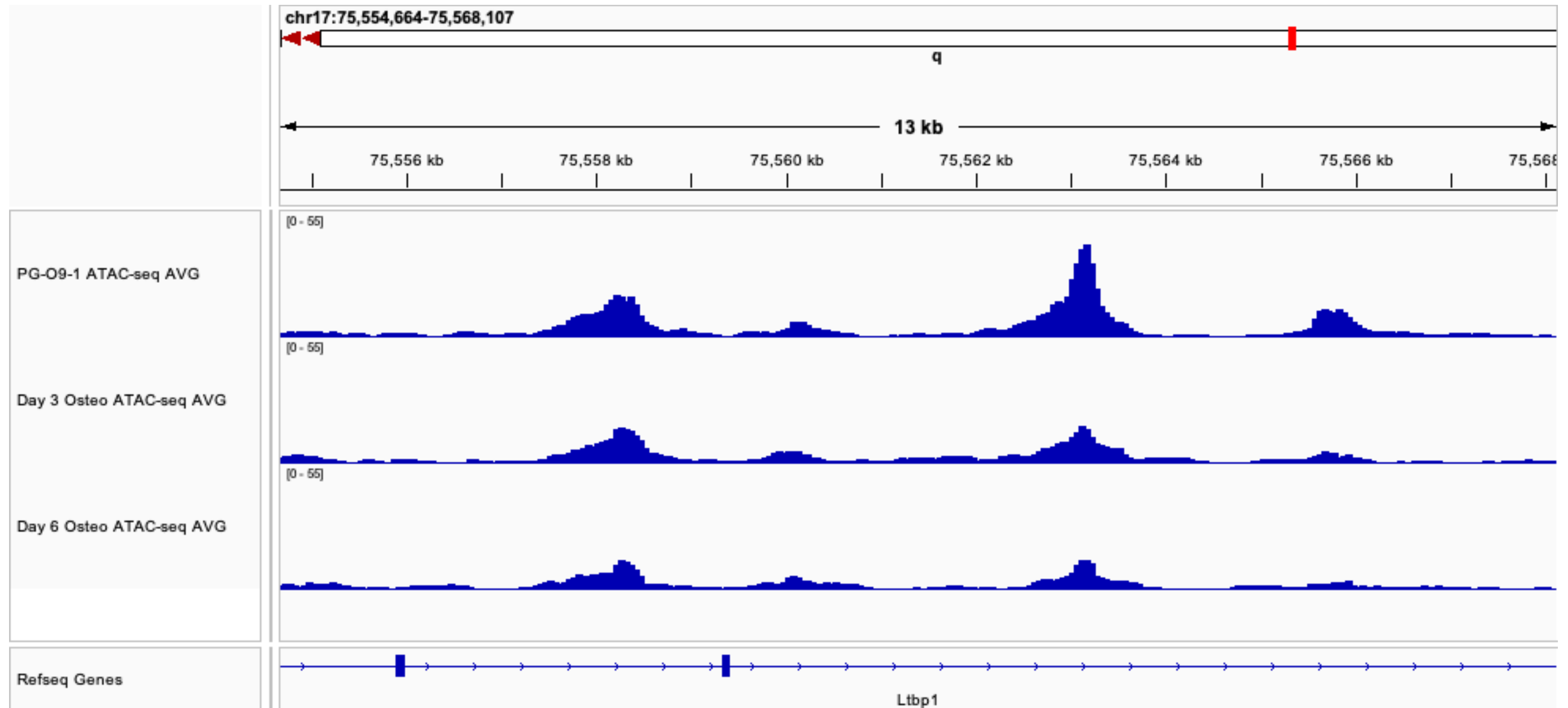
Differentially accessible peaks



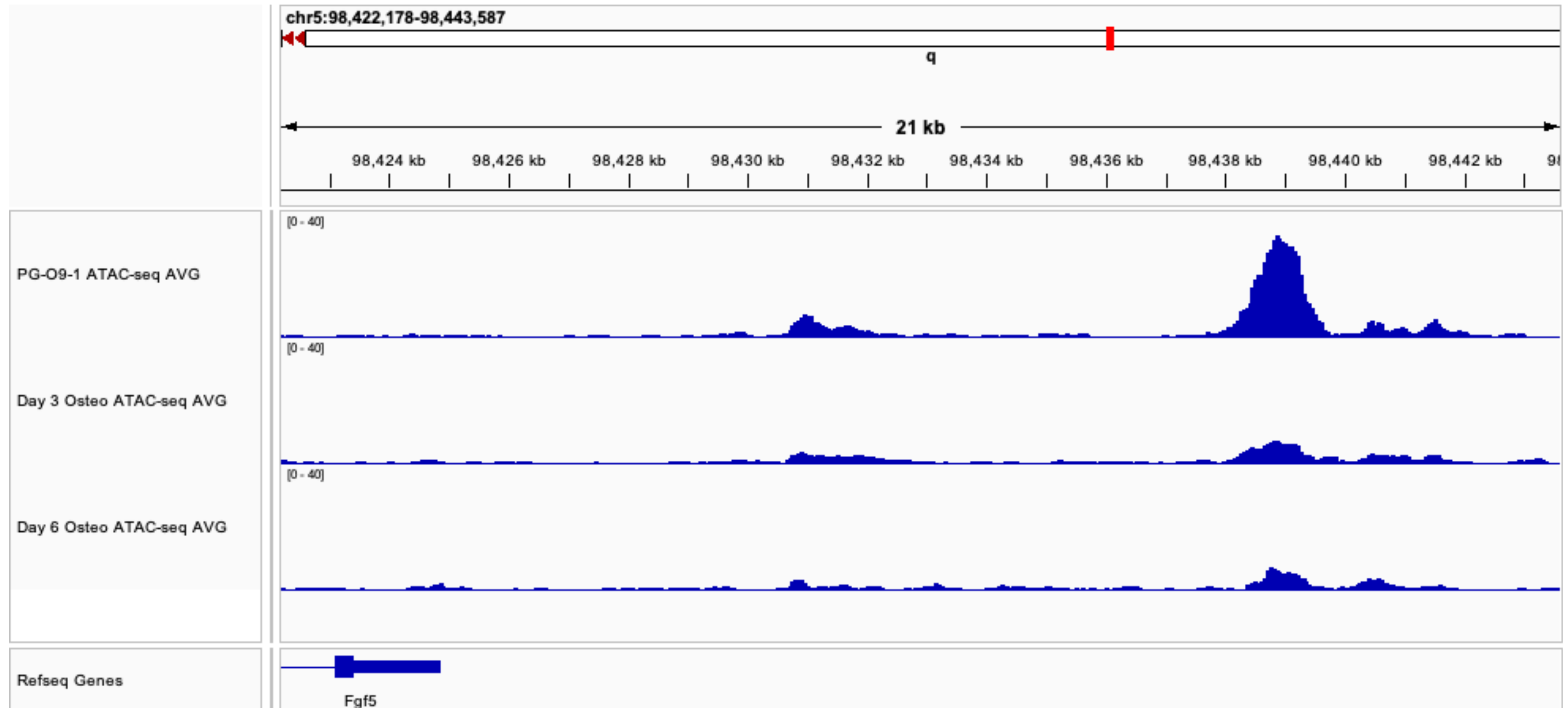
Differentially accessible peaks



Differentially accessible peaks



Differentially accessible peaks



Uneven MA plot

